

Edge-Optimized Deep Learning for Real-Time Crowd Density Estimation in Public Spaces

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
October 2025.

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Declaration

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

Accurate crowd density measurement is now very important for keeping our cities from becoming too crowded and causing disasters in mass gatherings. With large scale events such as mass gatherings in Bangladesh scheduled for July 2024, it is clear that the requirement for real-time and accurate data can no longer be met by conventional surveillance systems with tremendous latency introduction and severe compromise of privacy. There is a painful trade-off in existing deep learning methods: models that perform well and are accurate, such as CSRNet, are computationally too expensive for use in low-power edge devices, and on the other hand, models for efficient recognition of these devices, such as LCDNet, cannot map out in dense scenarios. This is an unsustainable "one-size-fits-all" point of view for dynamic and real-world environments. To bridge this gap, this thesis presents an adaptive and privacy-preserving hybrid architecture that applies the powerful combination of two specialized networks. Our system contains a MobileNetV2-based routing classifier that will evaluate the scene complexity in real-time. For sparse and simple scenes, it feeds the input to an efficient LCDNet-based model. For congested and complex scenes, it routes to a Knowledge-Distilled MobileCount model, a compact student network trained under supervision of a frozen CSRNet teacher, which is not deployed at inference. This dynamic selection mechanism ensures that computational resources are not wasted without making a sacrifice in accuracy when it is required most. Importantly, the whole process is designed for on-device execution on low-power hardware such as the NVIDIA Jetson Nano. By locally processing video and sending aggregated, anonymous density features, our framework upholds the privacy of individuals and significantly eases the bandwidth needed. This research, by conducting an extensive cross-database and empirical optimization study, shows how ethical smart surveillance is manageable in the next generation, so that both performance and public trust are optimal

Keywords: Crowd Density Estimation, Hybrid Architecture, Knowledge Distillation, Privacy-Preserving Surveillance, Edge Deployment, Real-Time Inference, Convolutional Neural Networks, MobileNetV2, LCDNet, MobileCount, CSRNet, Smart Surveillance.

Dedication (Optional)

A dedication is the expression of friendly connection or thanks by the author towards another person. It can occupy one or multiple lines depending on its importance. You can remove this page if you want.

Acknowledgement

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Chapter 1

Introduction

1.1 Background

Crowd density estimation has grown into a more important issue in terms of providing public safety, efficient urban administration, and event organization in contemporary societies. In the era of accelerated urbanization and increased rate of mega events like political rallies, religious events, concerts and sports matches, strong and scalable monitoring systems are now more valuable than ever before.

Such tragic cases as the overcrowding caused by stampede at big events prove the necessity of real-time, accurate measurements of the number of crowds. Prominent case in point is the July 2024 uprising in Bangladesh on 3 August, student coordinators announced a unified demand to overthrow the regime, resulting in mass meetings, which attracted strong international attention. In these incidents, journalists and observers relied on timely and dependable crowd estimates to get a sense of scale and risk to depict the social and humanitarian priority of reliable crowd surveillance instruments and which led to an estimate 5th August gathering and a regime fall.

Conventional surveillance is based on centralized CCTV systems, drones and powerful servers [1], [2]. These methods are useful but have serious drawbacks: they require high computational resources, have high latency, are not very scalable, and privacy is not liked by the population. Moreover, most current systems process each and every scene identically, wasting resources in low-density situations, and performing poorly in high-density ones. At the same time, the edge computing platforms (e.g., NVIDIA Jetson Nano) allow the introduction of distributed, low-latency, and privacy-preserving solutions but introduce severe memory, power, and computation limitations [3], [4]. This constellation of needs motivates research into hybrid, adaptive, and privacy-aware crowd density estimation suitable for edge deployment.

1.2 Rational of the Study or Motivation

Multiple observations and practical requirements provide the motivation to this study. First of all Accuracy vs. Efficiency Trade-off, where modern models such as CSRNet are very accurate in dense crowd scenarios to the point that they are computationally expensive and cannot be used in low-power edge devices [5]. Lightweight models like LCDNet are efficient on-device, but may not be accurate in dense or complicated scenes [6]. This gap can be closed with a hybrid approach that offers

precision in cases where it is required and efficiency in other cases. Crowd Density Estimation, Hybrid Architecture, Knowledge Distillation, Privacy-Preserving Surveillance, Edge Deployment, Real-Time Inference, Convolutional Neural Networks, MobileNetV2, LCDNet, MobileCount, CSRNet, Smart Surveillance. Apart from this Dynamic Scene Complexity, such as crowd scenes move through time and space- an event location can switch back and forth between thin and thick [7]. Such variability cannot be best handled by having a static single-model deployment [8]. Adaptive inference (direct inputs to alternate models) may save average computation, with accuracy being maintained during high density intervals [9], [10]. Moreover, Edge first privacy requirement, Transmission of raw video is an issue because of public awareness to surveillance and abuse of data. Edge Processing (on user devices or local nodes) and only transmitting aggregated density features can enhance privacy, as well as bandwidth [11], [12]. The given architecture clearly has on-device feature extraction as a priority. Furthermore Practical Deployment Constraints like Realistic deployments are focus on devices with strict memory and power constraints such as Jetson Nano [3]. Real world problems need more than new architectures, and practical model optimization such as pruning, quantization, inference acceleration and routing mechanisms that can be minimally energy consuming and latency sensitive [13], [14], [4]. With a total deployed pipeline well under 5 M parameters, it is easily achievable on resource constrained edge hardware. As well as that Gap in Current Literature, although accuracy-based and efficiency-based models are researched separately in the previous literature, the existing literature does not focus on the systematic study of learned routing with hybrid architectures to estimate the number of crowds in ultra-low-power edge settings [5], [6]. Such a gap implies that there is an apparent research opportunity to develop and test an adaptive hybrid system. Lastly, Societal Relevance for example certain situations such as the July 2024 uprising show the necessity of the availability of ethical and privacy-wise solutions that can provide an immediate information flow to responders, journalists, and organizers. The thesis is intended to make a contribution to a technically sound solution that has direct practical and social applicability. The work in this paper is motivated by these arguments, and therefore takes a hybrid, flexible, and privacy-friendly approach of using knowledge-distilled MobileCount accuracy for dense scenes, and LCDNet efficiency for sparse scenes, with a simple MobileNetV2 based routing classifier acting as the mediator and edge-friendly optimizations.

1.3 Problem Statement

In spite of the progress in deep learning in crowd counting, there are four endemic challenges in the practical implementation in the smart city and event environment: Statistical pipelines: These pipelines rely on a single model no matter the complexity of the scene, and result in either redundant computation or inaccurate performance.

1. Static pipelines: These pipelines rely on a single model no matter the complexity of the scene, and result in either redundant computation or inaccurate performance.
2. Accuracy-efficiency trade-off: precise models are too bulky to have on edge devices, whereas in dense crowds inaccurate models are lightweight.
3. Lack of adaptive routing: very little is done systematically to assess lightweight classifiers to route to a special density estimator.
4. Privacy risks: most of the systems require sending raw images to a central server,

which is ethically and legally questionable.

The thesis question is::How to develop an adaptive privacy preserving crowd density estimation system that can dynamically trade off between accuracy and efficiency and can be knowledge distilled to enhance lightweight specialists, and can be deployed on ultra-low-power edge hardware?

1.4 Objective

This study aims to develop an edge-minimizing, privacy-preserving hybrid deep learning architecture dynamically switching between LCDNet and Knowledge-Distilled MobileCount to estimate real-world crowd size in open spaces. Otherwise a lightweight scene classifier will be developed to classify scenes (such as sparse or dense), which will allow adaptive routing. Additionally, a dual-path setup will be adopted: a fast path based on LCDNet for sparse scenes and a slow path based on MobileCount for dense scenes will be implemented to switch based on the characteristics of the scene. The accuracy of the dense scene path will be enhanced by knowledge distillation from a frozen CSRNet teacher, while not increasing the inference cost. Furthermore, the readiness for edge deployment will be exemplified by the number of parameters, GFLOPs and latency profiling on both GPUs and CPUs, while quantization and TensorRT optimizations are set as future works. The framework is expected to be validated using the cross-dataset evaluation and ablation studies to examine the generalization capability, in addition to maintaining privacy using on-device processing and the transmission of aggregated density features only.

1.5 Methodology in Brief

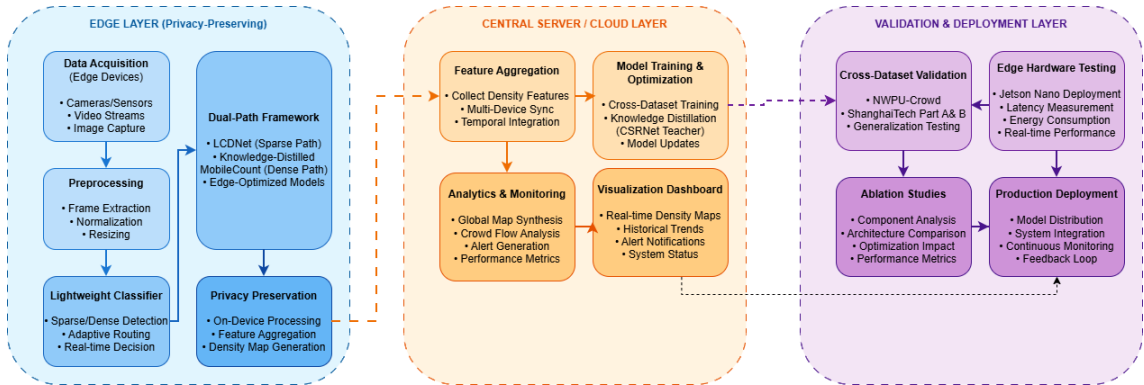


Figure 1.1: Proposed System Architecture for Privacy-Preserving Edge-Optimized Crowd Density Estimation

Figure 1.1: Proposed System Architecture for Privacy-Preserving Edge-Optimized Crowd Density Estimation

The proposed system is a hybrid system combining LCDNet with a Knowledge-Distilled MobileCount framework, which can get accurate, real-time and edge-optimized estimation of crowd density. This is started by data acquisition on distributed edge devices, with local processing of video frames to create compact density features, without the transfer of raw images so that individual privacy is not lost. A

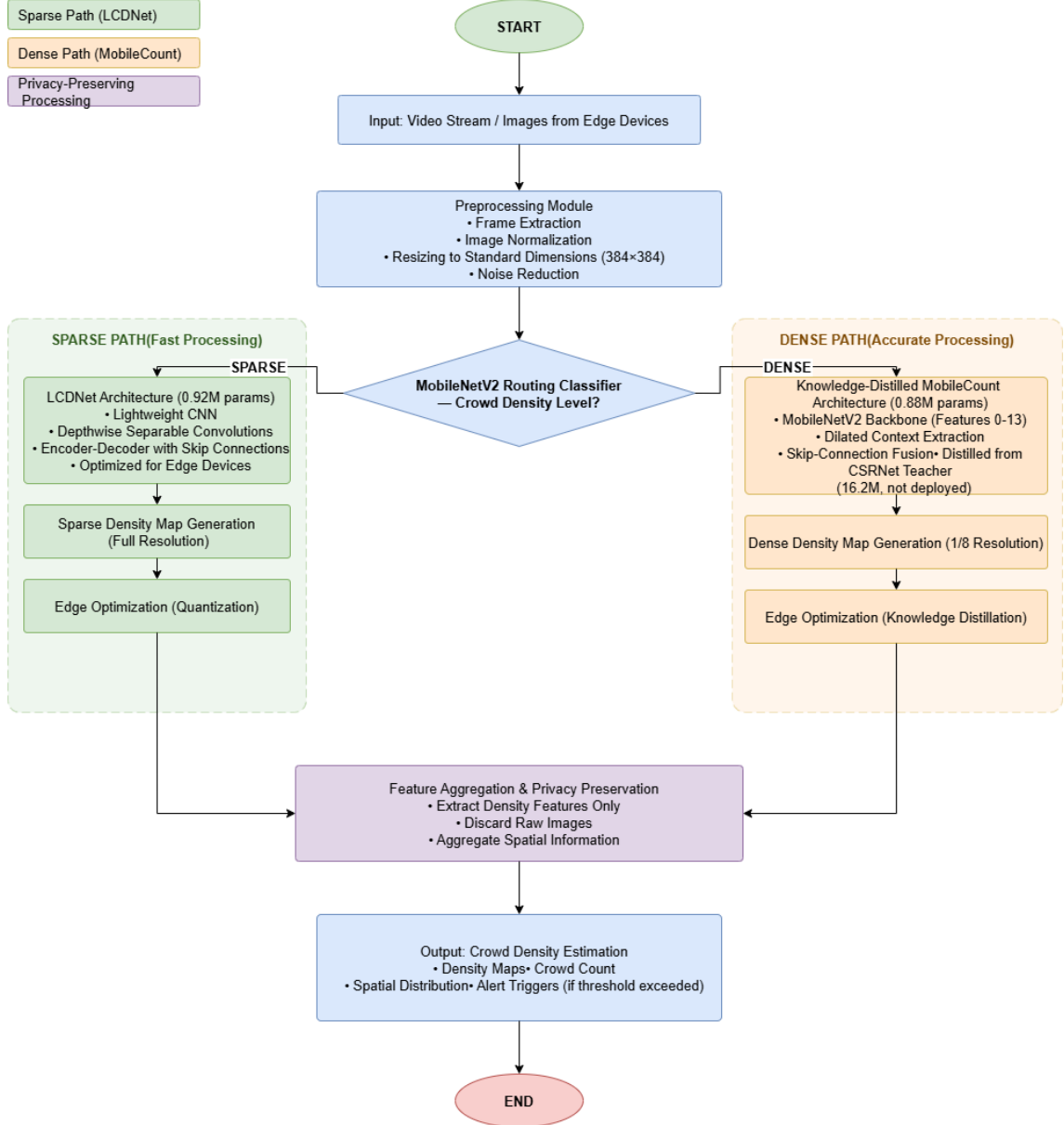


Figure 1.2: Workflow of the Proposed Hybrid Model

MobileNetV2-based routing classifier then real-time evaluates the scene complexity, and routes each input to one of two special models: sparse scenes are sent to the efficient LCDNet model, while dense scenes are sent to the model trained with the supervision of a frozen CSRNet teacher, which is called Knowledge-Distilled MobileCount model. This adaptive dual-path mechanism helps to combine the resulting density features, enhancing spatial accuracy and efficiency. To achieve the generalization ability across different scenes, the hybrid model is trained and validated on several benchmark datasets such as ShanghaiTech Part A, ShanghaiTech Part B and NWPU-Crowd. The system is also tested on edge hardware like the Jetson Nano to assess the real time performance, energy consumption and inference latency in realistic conditions.

1.6 Scopes and Challenges

The focus of this work is to develop a learned routing-based adaptive dual-path crowd density estimation system (LCDNet–Knowledge-Distilled MobileCount hybrid). The goal is to deploy and test on realistic low-power hardware, like the Jetson Nano. The system will be designed to respect privacy by processing and discarding the raw images locally, and sending only aggregated density features. To verify the generalization capability, cross-dataset benchmarks such as ShanghaiTech Part A, ShanghaiTech Part B and NWPU-Crowd are applied. Other issues that are not discussed in this work include large-scale long-term crowd-flow prediction, face recognition, identity tracking, and fine-grained behavioral analytics, for which only instantaneous density estimation is discussed. The key challenge is to have good counting accuracy within tight memory and power budgets with inexpensive, ubiquitous edge devices.

Chapter 2

Literature Review

2.1 Preliminaries

2.1.1 Crowd Density Estimation Fundamentals

Crowd density estimation aims at estimating the number of individuals in an image or a video frame without the need for explicitly detecting and tracking each person. However, with increasing crowd density the traditional object detection approach often fails because of severe occlusion, perspective change and overlap of human faces. In recent years, however, there has been a growing interest in density map regression methods that learn a mapping between an input image and a density map of a crowd, which yields an estimate of the number of people in the image. Instead of using bounding boxes around individuals, density map regression methods use a density map to estimate the total number of people whose integral is an approximation to the total number of people. In dense and highly occluded environments, the strategy has been shown to excel, especially where traditional detection-based pipelines may struggle to achieve good performance. In such cases, individual heads might not be identifiable, but the distribution of the density values of the image still gives enough information to obtain a good estimate of the size of the crowd. In order to generate these density maps, each location of the heads is generally convolved with a Gaussian kernel, resulting in a continuous representation of crowd distribution. The total number of people can be found by integrating over this density map. On the basis of this concept, in highly populated scenes, the performance of techniques based on regression is superior to detection-based ones [15]. This improvement is mostly attributed to the fact that regression methods do not require instance-level localization in very occluded cases which makes them more reliable in the more complicated environments for counting people in the crowd.

2.1.2 Convolutional Neural Networks for Density Estimation

Deep Convolutional Network is used to extract information hierarchically from the picture. This method has end to end learning capabilities. CSRNet is an example of this improvement [5]. It is associated with dilated convolution and uses VGG-16 in it. This method can capture both the pattern of the whole crowd and small person features as well as no spatial resolution is lost. On benchmark datasets like

UCF-QNRF, WorldExpo'10 and ShanghaiTech, this approach achieved 30-40% improvement in MAE over earlier methods [5]. On the other hand LCDNet is designed to use depthwise separable convolutions and pruning of unnecessary channels [16]. By dividing the into pointwise and depthwise operations LCDNet achieves to cut parameter count by 5-8 times without losing much accuracy means maintaining high accuracy in sparse and moderately density environments. By focusing efficiency and speed, LCDNet can run in real time on edge devices achieving exceeding more than 20 FPS on platforms such as Jetson Nano [16]. Moreover, an improved MobileCount is designed with a pre-trained MobileNetV2 backbone, dilated convolution block and skip-connection fusion block [17]. This architecture is even more appropriate for resource constrained environments as it generates density maps at a lower spatial resolution. Furthermore, using a larger teacher network, with knowledge distillation, the MobileCount can boost its density estimation ability without greater deployment complexity. As such, it is a potential solution for dense-scene specialization in hybrid routing approaches [17].

2.1.3 Edge Computing Architectures

New crowd monitoring applications are becoming more real-time based, and so are the platforms that will have to be used to deploy these systems. Hence, edge computing has proven to be a realistic option since it enables data to be processed at the source, instead of sending raw video streams to central servers. Besides these it can both decrease the latency and enhance privacy and bandwidth efficiency. There are various hardware platforms that have been created to enable edge deep learning applications, such as NVIDIA Jetson Nano, Jetson Xavier NX, Jetson Orin Nano, Google Coral TPU, and Intel Neural Compute Stick. When evaluating such devices, it was pointed out that they can be significantly different in terms of performance, depending on the architecture of the CNN used, whether it is a standard, desktop-oriented architecture or one specifically optimized for embedded systems [3]. In crowd counting applications, devices are likely to have limited memory, power budgets and computational resources. To boost the deployment efficiency of these networks, researchers have increasingly turned to lightweight backbones, structured pruning, and low-precision inference methods like INT8 quantization [4], [13]. Furthermore, tooling like NVIDIA TensorRT has been an integral part of any edge deployment. These frameworks provide a way to transform more complex architectures for crowd counting into efficient surrogates that can easily run on a constrained device without too much computational overhead. Therefore, edge computing systems can serve as a promising solution for implementing real-time, privacy-preserving, and resource-efficient crowd density estimation systems.

2.2 Review of Existing Research

2.2.1 Traditional Surveillance and Monitoring Systems

Initial crowd analysis systems were based on static CCTV networks or overhead cameras or aerial platforms like drones and on traditional computer vision methods or manual annotations. Those systems were the first and so far only attempts to automate crowd monitoring and estimation in surveillance scenarios. The vari-

ous methods of early density estimation and crowd-counting, which include background subtraction, feature-based regression, and simple blob analysis techniques [18]. These approaches worked well in low-to-moderate density settings with relatively well-controlled viewing conditions; however, they were not very successful when dealing with high occlusion levels, complex motion patterns, or significant changes in lighting. Given some of the drawbacks of the traditional surveillance methods, there was an increased interest in research on drone-based monitoring methods. When it comes to crowd monitoring, drones provide versatile perspectives and access to expansive areas that makes them appealing for such tasks. Yet, despite its benefits, drone-based systems also have a number of issues to consider, such as battery life, stability, and regulatory regulations. Moreover, traditional monitoring methods, such as those with fixed cameras and aerial cameras, have relied on centralized monitoring systems for data analysis, which raises issues of latency, bandwidth usage, and privacy. As the size and complexity of crowd monitoring applications grew, it became obvious that the drawbacks of the hand-crafted approach could no longer be contained. The success of such deep learning models in highly congested scenes is crucial and argued that the traditional feature-engineering methods are not suitable in modern large-scale deployments [19]. Researchers studied recent advances in neural architecture search and privacy-preserving learning architectures, and they found that federated learning holds a potential solution for mitigating privacy threats in surveillance and biometric applications [20]. All these studies argue that there is a gradually moving shift from the usual computer vision methods to intelligent and privacy-preserving deep learning systems.

2.2.2 Density Map Regression Approaches

CSRNet targets very dense congested crowd scenarios. By employing VGG-16 as a base, it uses dilated convolutions replacing pooling layers, maintaining spatial information and capturing larger fields. This convolution pattern uses rates in an exponential pattern 1, 2, 4 and 8, allowing the network to capture both broader crowd pattern and small scale crowd structures. CSRNet reached MAE of 68.2 on ShanghaiTech, marking 38% improvement over MCNN. The architecture works well in congested scenarios by preserving spatial information and a wide contextual patterns. However CSRNet cannot directly deployed on ultra limited resource edge devices without major adjustments. However, CSRNet is still very compute-intensive and not directly applicable to the edge computing devices with very limited resources. Thus, recent research has focused on the use of CSRNet as a teacher for knowledge distillation instead of a deployment model. In this context, the knowledge acquired by CSRNet is learned by a lightweight student network which can then be deployed while retaining a lot of the original model’s accuracy [5], [17], [21].

2.2.3 Lightweight Architectures for Edge Deployment

Although the accuracy of counting was found to be significantly better using models based on density regression, such models are often computationally too demanding to be used on embedded platforms. Thus, novel lightweight crowd counting architectures that are optimised for edge environments were conceived and developed. One family of such methods are the so-called lightweight density estimation networks,

such as LCDNet, that use depthwise separable convolutions and channel pruning to decrease the number of parameters and the computational load [6]. Representative models usually are encoder-decoder models with skip connections and depthwise separable blocks from MobileNet and Xception. Such networks can achieve a significantly lower number of parameters than heavier networks, and competitive MAE on sparse-to-moderate crowd scenes, by utilizing the efficient design principles [16]. But light architectures degrade in the case of high congestion. Receptive fields are generally larger and feature representations are more extensive when the scene is dense, and it is often necessary to differentiate small variations in the crowd density. This results in a trade-off between computing efficiency and accuracy of the estimation. In order to tackle this difficulty, Hybrid architectures were proposed and advocated to dynamically allocate computational resources based on the complexity of the scene. In these systems, the light-weight models are applied to sparse scenes, and more powerful models are used for dense scenes. The adaptive strategy is to find an optimum balance between speed and accuracy [17]. The MobileNet-based architectures are another significant approach in the field of lightweight crowd counting. Due to the low computation complexity and compatibility to mobile hardware, several studies have used MobileNetV1 and MobileNetV2 backbones. To capture multi-scale contextual information, dilated convolutional blocks in the decoder and MobileNetV2 as an encoder [22]. The resulting MobileNet-Count architecture could cut the number of parameters in the network down to just a few million and still achieve competitive results on the MobileNet-Count benchmark.

2.2.4 Adaptive and Attention-Based Mechanisms

Another important research stream has been towards adaptive and attention-based mechanisms that dynamically allocate computational resources based on scene characteristics, beyond lightweight architectures. These methods try to enhance their accuracy by selectively focusing on the more informative regions instead of considering all regions equally. SANet dynamically emphasizes features from different scales for crowd counting. The scale-aware attention module selectively highlights relevant scale based features and improves performance with highly variable crowd scenes. SANet achieved an MAE of 67.0 showing a 15% improvement than fixed weight fusion models on WorldExpo'10 benchmark. Attention improves crowd counting models but this can be a problem for computation and harder to run on lightweight edge devices. Attention designs that maintain performance accuracy while ensuring they can run on resource limited devices are emphasized [16], [23]. PAN manages scale variations from different camera angles. PAN predicted perspective maps to adjust its features to correct perspective distortions and resulting in more accurate crowd density estimation. This approach shows that this method achieves accurate results than the methods that ignore it. Even though this method depends on available perspective data, it paves the way for a strong foundation for future work on hybrid approaches that are capable of addressing scale variations [24]. The AMR network segments an image and employs specialized regression functions for each region. AMR can address diverse crowd distributions within a single image. This network relies on soft routing weighted routing rather than hard routing model. Our approach takes the idea further by routing whole input images to the appropriate specialized network. AMR outperformed CSRNet showing MAE gains of 8-12%

without extra computational cost [25].

2.2.5 Hybrid and Ensemble Approaches

However, while researchers are still seeking more accurate counting techniques, another option that sparked interest was hybrid and ensemble methods. Instead of using just one network, these techniques use more than one network or more than one processing stage to increasingly improve the density estimate and this influential model is called ic-CNN [26]. The structure is a series of networks that refine the density maps one after the other. In the first step an initial coarse density estimate is produced, followed by further steps that use both the original image and past predictions to produce a more accurate density estimate. This progressive refinement method is much more effective on spatial detail and counting accuracy than the single-pass method. But, the need for several forward passes means that the inference time and computational resources are also costly. This approach is especially of great interest in the case of extreme perspective variation and high traffic density. However, the multi-stage approach adds further computation time requirements which can be undesirable in real-time applications, as in the case of ic-CNN. Efficiencies of repeated processing stages have led researchers to consider alternative hybrid approaches that are based on hard routing. In contrast to iterative refinement systems, hard-routing systems use a light classifier to determine for each image of an input image set which specialist model to route the image to. Thus, only one density estimator is run for each image reducing considerably the amount of computational resources needed. This not only avoids forward passes as found in cascaded architectures but also offers model specialization. In this context, hard routing has been a promising alternative solution for latency-critical edge deployments, where computing resources are limited [8].

2.2.6 Model Optimization for Edge Deployment

Crowd counting is becoming more complicated with the development of more sophisticated models, and this is becoming a challenge when deployed at resource constrained edge devices. As a result, researchers have investigated different ways to optimize the models and decrease the amount of computation required while retaining accuracy. Some of the most popular methods are quantization, pruning, and knowledge distillation.

Quantization

One of the best ways to optimize deployment is to use quantization, which decreases the precision of the numbers in the model from FP32 to INT8, for instance. Model size can be further reduced and inference can be accelerated by post-training quantization or quantization aware training. NVIDIA’s TensorRT and other optimization frameworks analyze activation distributions to decide on the scaling factors, and typically get a 2-4x speedup while barely denting performance in vision-related tasks [3], [13]. Furthermore, density map pixels can be perturbed slightly, as density maps are aggregated to provide final counts of the crowd. In addition, crowd counting models are also fairly tolerant of small perturbations in density map pixels as the

final crowd estimates are derived from the density map. This means that, generally, quantization affects the counting accuracy by a small amount. But not all architectures are equal when it comes to low-precision. However, attention-heavy networks and very shallow models can be less tolerant of quantization noise and require attention to parameters such as the quantization, or bin size, to ensure correct deployment.

Pruning

Pruning is a technique designed to prune any parameters that don't significantly affect the performance of the model, whereas quantization is designed to be as numerically precise as possible. This can be done either by unstructured sparsity (where individual weights are removed) or structured sparsity (where entire filters, channels or blocks are removed). A gradient-based approach here estimates how important a filter is, by calculating its removal impact on the loss function [14]. In this way, significant compression of the model can be obtained. However, the benefit that can be gained from pruning will greatly vary depending upon the pruning system used. Unstructured pruning can lead to high compression ratios but sparse networks are not necessarily fast to be achieved unless they are used with specific optimized kernels. On the other hand, structured pruning can directly lower computation and memory bandwidth cost to make it more ready for being deployed on the contemporary edge GPUs.

Knowledge Distillation

Along with the quantization and pruning strategies, there have been studies on transferring knowledge from large models to smaller ones. In the approach of knowledge distillation (KD), a small student network is trained to imitate the soft predictions or intermediate representations of a larger teacher network [21]. To this end, study proposed to transfer knowledge from a heavy-weighted network like CSRNet and VGG-based network to a light-weighted network like MobileNet- or ShuffleNet-based students via knowledge distillation to the crowd counting task [17]. Their results showed that significant reductions in the number of parameters and inference latency can be obtained at competitive accuracy. Moreover, knowledge distillation is more effective when performed in a sequence optimization pipeline along with quantization and pruning. This approach can allow for high compression yet still maintain the key information that the original network has acquired. In crowd counting applications, a powerful model like CSRNet can then be used as a "frozen teacher" that is used to transfer crowd-density knowledge to a lightweight "student network", which is then kept for deployment after discarding the teacher network [17].

2.2.7 Cross-Dataset Generalization Studies

Domain Shift and Adaptation

While a lot of crowd counting models have great benchmark performance, it is still a big challenge to generalize in different datasets. To address this question,

trained models on ShanghaiTech and evaluated on UCF-QNRF, where noted significant performance drops in response to the variations in scene composition, camera viewpoints, and crowd characteristics, as well as to different labeling conventions [27]. In order to solve the problem, Gao et al. proposed to investigate adversarial domain adaptation methods which minimize the difference in feature distributions between source domain and target domain with a discriminator network. Though this methodology mitigated generalization error, it did not completely remove it. The results of these studies thus demonstrate the common overfitting of the specific features of the datasets, thus reinforcing the importance of more powerful feature representations and more effective adaptation strategies.

Multi-Dataset Training

Because of the lack of data to rely on just one set of data for training, researchers have been urging the use of more data sources. Training can be done on heterogeneous data sets to enhance the robustness of the model, as well as to lessen the sensitivity of the model to domain shifts [28], [29]. One important contribution along this line is the NWPU-Crowd dataset that provides thousands of images of various density, environment, and point of view [30]. NWPU-Crowd offers a more diverse benchmark than previous work, making it a useful tool for assessing counting performance and cross-dataset generalization performance.

2.2.8 Privacy-Preserving Crowd Monitoring

Federated Learning

A promising approach is federated learning, which allows edge devices to train local models with private video data and only send updates to the model, rather than raw video data. The challenges do not come without a price, however, since federated learning is a different kind of challenge. When there are many participating devices, there can be a lot of communication overhead, and gradients themselves can leak sensitive data. To tackle this concern, an idea is that client-level differential privacy can minimize gradient leakage by carefully adding noise to the updates of the model. However, this extra privacy often comes at the expense of model accuracy [11].

On-Device Processing and Local Privacy Mechanisms

In addition to federated learning, researchers have also examined approaches that utilize on-device data processing without moving it to any other location. Study showed the privacy-preserving crowd monitoring methods via local differential privacy and compact sketches like Bloom filters and the methods combine crowd information and only expose non-identifiable personal information [12]. Importantly, such strategies fit well with a number of regulatory frameworks, such as GDPR and emerging privacy frameworks. But greater privacy protections may come with a loss of data usefulness and analytical accuracy. This trade-off has inspired a range of architectures to process images locally, and to only send annotations that represent aggregates of the images, such as low-resolution density maps, which reduces privacy concerns and bandwidth needs.

2.2.9 Routing and Model Selection Mechanism

Learned Routing in Neural Networks

With the growing specialisation of models, the question as to which of them should act upon a specific input becomes crucial. To overcome this problem, developed conditional computation and mixture-of-experts architectures, in which their inputs are dynamically directed to specific expert models through a gating network [8]. By selectively activating only a certain number of experts, these systems could lower the average number of computations and still achieve a high accuracy. On this basis, adaptive inference graphs, which adaptively skip residual blocks under learnt policies [9]. In parallel, SkipNet uses reinforcement learning to decide whether to skip or not skip layers for a specific input [10]. These studies illustrate adaptive computation’s efficacy in improving efficiency together. Crowd counting can be applied to this with a light weight image classifier trained to distinguish between sparse and dense scenes and then classify each image to the appropriate density estimator. This binary routing strategy is a simpler and yet very effective method for scene-aware computational allocation, in comparison to reinforcement-learning-based methods.

Scene Complexity Classification

The scene complexity estimation is closely related with routing. The study showed the characteristics of the edge density and motion information and proposed a scene complexity classification for video coding [31]. They divided the scenes into simple, medium and complex groups, which facilitated efficient allocation of the computational resources in their system [31]. The same approach applies to crowd counting: the complexity of a scene can be estimated and then either a lightweight counting model or a heavyweight model can be run depending on the estimation. Therefore, complexity classification is an important fundamental for adaptive crowd counting systems.

2.2.10 Real-World Deployment Case Studies

Smart City Deployments

Most of the crowd counting techniques are tested in experimental environments while deploying in the real world adds challenges. Places like transportation stations, shopping zones and other urban areas are populated with high population densities [19]. In such environments, CNN-based crowd counting systems can be used in various applications such as public safety monitoring, traffic management, and event coordination. But deployments in the real world have to deal with different camera quality, weather conditions, limited network connectivity and low latency requirements. These are some of the challenges that have further entrenched interest in edge-based processing architectures.

Event-Specific Monitoring

In addition to city-wide deployments, there are also temporary large scale events where crowd estimation is of critical importance. The passive RF sensor network is for crowd monitoring in public gatherings [1]. Unlike vision-based systems, RF

sensing can ensure a satisfactory crowd estimation accuracy while, at the same time, avoiding a variety of privacy problems. However, it is not possible to obtain detailed spatial density maps like those obtained with camera-based approaches using RF approaches. Nevertheless, they show that there are specific cases where other sensing modalities can be used in conjunction with a vision system.

2.3 Summary of Key Findings

The vast body of work on crowd counting and density estimation highlights several ongoing challenges, which are still impeding the practical use. Firstly, there is an ultra-low power deployment gap, as many good models like CSRNet, SANet, etc. were specifically built for powerful GPUs and are not optimised for platforms like Jetson Nano without a lot of modifications. While lightweight MobileNet-based architectures are more efficient, they tend to be less accurate in very dense scenes [3], [5], [6]. Secondly, there is a cross-dataset generalization gap that is still apparent. The majority of benchmark datasets are dedicated to certain geographic and cultural settings, and models that are trained on one set of data often fail to perform well on another one because of the difference in context or domain shift in the data sets. In recent years, deep learning methods such as domain adaptation and multi-dataset training have yielded promising results, but strong generalization is still a challenge. Thirdly, an adaptive inference gap can be observed. In related computer vision applications, adaptive routing and mixture-of-experts architectures have been successful in achieving remarkable efficiency gains. These, however, have limited applications as far as crowd counting is concerned, even though they may be useful to distribute computation as a function of scene complexity.

Moreover, there is a gap in the holistic optimization; quantization, pruning, and knowledge distillation have been studied independently instead of being considered as optimization pipelines. However, little guidance exists on how these methods can be optimally combined to achieve speed, power and accuracy. Methods like federated learning and differential privacy help mitigate the need for centralised data collection, but can also create communication costs and potential loss of performance [11], [12]. When combined, these gaps highlight the challenge of realizing a crowd counting model developed in the lab to a realistic, resource-limited application. To tackle these issues, this research suggests a dual-path architecture comprising a lightweight sparse specialist, a knowledge-distilled dense specialist, and a binary routing classifier, which is specially designed to be used on ultra-low-power edge devices [6], [17], [21].

““

Chapter 3

Societal Impact & Constraints

3.1 Final Specifications and Requirements

3.1.1 System Overview

In this chapter, the final specifications and engineering requirements of the proposed hybrid crowd density estimation system are defined. The system is designed to be lightweight, privacy-preserving, and edge-deployable, which involves a learned routing mechanism to dynamically route the input image to one of two specialist density estimators: LCDNet for sparse scenes, and a knowledge-distilled MobileCount model for dense scenes. The routing module is a binary classifier using MobileNetV2. The full pipeline is entirely on the edge device and not reliant on a constant connection to the cloud.

3.1.2 Functional Requirements

The system must fulfil the following functional requirements:

Real-Time Crowd Density Estimation: The system should be able to take video frames, generate per-pixel density maps, and calculate a density estimate from them. Processing should take place at a rate appropriate for near real time monitoring.

Adaptive Scene Routing: An input image will be classified by a routing classifier into a sparse or dense crowd scene and passed on to the corresponding density estimator. The route will be determined automatically.

Dual-Path Density Estimation: Two lightweight density estimation models will be kept: one for sparse scenes (LCDNet) and another for dense scenes (MobileCount, improved by knowledge distillation from CSRNet). During inference, only one model is to be run per image.

Privacy-Preserving Output: This system shall not capture, transmit or process any biometrically identifiable features, but shall output density maps that capture the spatial distribution and estimated number of individuals.

Edge Deployment Compatibility: All the pipeline components should run on edge hardware (such as NVIDIA Jetson Nano) during inference, without needing cloud offloading.

Cross-Dataset Generalisability: The system should perform well in a variety of crowd scenarios and different benchmark datasets, without the need for significant retraining on the specific dataset.

3.1.3 Non-Functional Requirements

Computational Efficiency: The total deployed footprint of the three model parts (router, LCDNet, MobileCount) shall be minimised in order to fit into the memory and power limitations of the ultra-low-power edge hardware.

Modularity: The system architecture shall be modular, allowing for the replacement or update of the router, sparse specialist or dense specialist without re-training the whole pipeline.

Reproducibility: All training scripts, configuration files and preprocessing routines will be version controlled and documented to ensure complete reproducibility of pipeline.

Regulatory Compliance: It should be designed to comply with applicable data-protection regulations, such as the General Data Protection Regulation (GDPR) [32] to ensure that no personal data is processed and stored outside the context of the density map generation.

Maintainability: The codebase shall be structured to be easily extendible and maintainable: extensions like quantisation, specialised optimisations for various hardware, further model variants etc. shall be possible.

3.1.4 Dataset and Training Requirements

The system will be trained and primarily tested using the NWPU-Crowd dataset [30] that contains a large and diverse set of crowd annotated images with varying levels of scene density. Generalisation will be tested on the ShanghaiTech Parts A and B [15] with a cross-dataset evaluation. The ground-truth density maps are obtained using adaptive Gaussian kernels that are created based on the distances between annotated head points using k-nearest-neighbour distance distribution as discussed by Zhang [15]. Images from all datasets are resized to a standard size (384×384) with standard ImageNet normalisation applied throughout for density estimation and 224×224 for the routing classifier.

3.1.5 Hardware Target

The main deployment target is the NVIDIA Jetson Nano, a single-board computer featuring a 128-core Maxwell GPU, quad-core ARM Cortex-A57 CPU and 4 GB of LPDDR4 memory with a power envelope of 5–10 W [33]. This is the intended deployment platform for this system, and the following performance figures are based on the NVIDIA GeForce RTX 3050 GPU and single-thread CPU, as future validation will be performed on the Jetson Nano. Other hardware designs under consideration for the wider design are Raspberry Pi and similar single-board computers with CPU only architectures.

3.2 Societal Impact

The need of the hour is to address the crucial gap in crowd monitoring systems: the lack of accurate, real-time crowd density analysis without expensive cloud infrastructure. The proposed system enables intelligent crowd monitoring directly on edge devices, making it accessible to resource-constrained organisations. In particular, its societal impact is reflected in three key areas: enhanced public safety and emergency response, improved data sovereignty and privacy preservation, and greater accessibility and digital equity.

Public Safety and Emergency Response: The traditional crowd monitoring systems are based on centralized cloud computing, which face latency issues, privacy risks and need robust internet connectivity. In contrast, the proposed hybrid edge-based system removes such dependencies: all inference is done on edge, eliminating the need for cloud servers. As a result, this design is especially well suited for high-stakes public safety applications. For example, in the case of concert halls, sports stadiums and festivals, the system can be installed on the edge of the network to track the density of people in real time and alert people before the venue reaches unsafe levels. Furthermore, the dual-path routing design guarantees that both sparse and dense crowd conditions are dealt with appropriately by the most suitable specialist model. In addition, rapid crowd density assessment is a major factor in coordinating evacuation in emergency situations. The edge-based design will allow it to continue functioning if the network infrastructure is compromised and avoid relying on cloud connectivity during critical situations. Consequently, the resulting spatial density maps are useful to give actionable spatial information for evacuation flows. Beyond emergency response, municipal governments have the ability to use these to enhance the design of public transport, pedestrian infrastructure, and allocation of security resources. Moreover, this portable architecture with the edge allows for the cost-efficient installation of the system in various public areas without the need to invest in dedicated infrastructure.

Data Sovereignty and Privacy Preservation: The system’s edge-first architecture directly responds to the increasing societal concerns about surveillance, data sovereignty and personal privacy. The system does not create identifiable data records by processing all video data locally on the edge device and only sending density maps of the images, not video data streams. By design, the density map outputs do not retain or disclose any personal biometric data but only statistical information on the spatial distribution of crowd presence, making the system GDPR-compliant in adherence to the GDPR principles of data minimisation and purpose limitation [32]. Consequently, this privacy-enhancing tool is especially useful in sensitive deployment environments. For example, healthcare facilities can use the system to monitor occupancy and patient flow patterns in waiting rooms without capturing identifiable faces, helping to meet healthcare data protection requirements such as HIPAA while supporting more equitable resource allocation. Similarly, schools and universities can monitor campus occupancy to improve safety without violating the privacy rights of students and staff, while municipal authorities can collect crowd analytics data to support urban planning decisions without creating

large-scale, high-risk surveillance databases.

Accessibility and Digital Equity: The hybrid architecture increases the scope of sophisticated crowd monitoring for organisations which otherwise would be cost-restricted. The lightweight deployed models enable the system to be deployed on commercially available edge devices rather than high-end GPU servers or subscription services to the cloud. As a result, occupancy monitoring can be used for safety management at small venues, such as community centers, places of worship and small businesses, without significant capital investment. Likewise, intelligent crowd monitoring is possible in developing regions without requiring high-speed internet connections or cloud computing, thereby avoiding the high costs of infrastructure and connectivity. Furthermore, the open-source codebase and modular design offer a practical platform for education in computer vision, deep learning, and edge deployment principles.

3.3 Environmental Impact

Recently, the resource utility of deep learning models has been a hot topic of discussion on the environmental impact of AI systems [34]. Training at scale is energy consuming and inference at scale makes a significant contribution to cumulative carbon emissions. The proposed system’s architectural efficiency and adaptive resource management provide a meaningful answer to these questions.

3.3.1 Computational and Energy Efficiency

The hybrid routing strategy offers fewer average computational costs than if a single large model were used for each object. The primary efficiency improvement is due to two synergistic effects:

Model Efficiency: The memory and computation usage of the deployed density estimators (LCDNet and the knowledge-distilled MobileCount) are comparable and significantly less than that of their heavier counterparts, like CSRNet. LCDNet does this by decomposing the standard convolution into a depth-wise and pointwise component, that are far fewer in number of parameters and arithmetic operations, respectively [35]. MobileCount also relies on MobileNetV2’s inverted residual structure and linear bottlenecks [36] to reduce computations.

Adaptive Computation: The routing mechanism ensures that computationally lighter inference is applied to scenes that do not require heavier processing. Sparse scenes, which constitute a meaningful proportion of real-world deployments, are handled by LCDNet, which is architecturally less demanding than MobileCount. This avoids the energy overhead of unnecessarily running the denser model uniformly across all inputs.

Memory Bandwidth Reduction: Smaller model sizes reduce the volume of memory transfers required during inference. Since memory access is a primary driver of energy consumption in edge GPU architectures [37], compactness directly translates to more energy-efficient operation.

3.3.2 Training and Lifecycle Considerations

The efficiency of the training pipeline has been designed. Instead of fine-tuning each component from scratch, the system uses the knowledge distillation technique, using the dense-scene specialist (MobileCount) to be initialized with a pretrained checkpoint and fine-tuned with limited epochs of training based on the outputs of the CSRNet as auxiliary supervision targets. This warm start approach significantly decreases the training computation as compared with training from random initialisation. After distillation, the teacher network (CSRNet) is removed, and will no longer participate in the deployed pipeline, thus keeping the running footprint lightweight.

3.3.3 Quantitative Environmental Accountability

To align with the principles of responsible AI development [38] this system’s environmental impact is partially characterised using FLOPs measurements. Per-device energy consumption on target edge hardware is identified as future work. The following measurements are planned or currently under way:

Energy per Inference: Measurement of per-inference energy usage on the sparse and dense paths on target edge hardware, to quantify the marginal cost of routing.

Carbon Footprint Estimation: Measurement of energy consumption and the use of local factors (carbon intensity factors) to estimate the CO_2 equivalents per inference cycle.

Comparative Assessment: Comparing the energy profile of the hybrid system with single-model baselines that are applied consistently to the various models, to measure the environmental benefit of routing.

Lifecycle Assessment: based on training energy, deployment facility overheads, and operational energy for a typical deployment time span, as suggested by Bannour [39].

3.4 Ethical Issues

The use of automated crowd monitoring systems brings with it a number of ethical issues that need to be taken into consideration at the design stage as well as in the operating procedures for the use of the system. This section outlines the key ethical issues that are pertinent to the proposed system and the decisions taken to address these issues to reduce the risks involved.

The most important ethical question in crowd monitoring is whether technology can enable disproportionate surveillance. Automated systems that continuously observe people in public or semi-public spaces generate power that may be used for monitoring, tracking, or profiling individuals and groups beyond their intended purpose. The proposed system addresses this concern through privacy-preserving architecture, performing all inference locally on edge hardware without sending or storing raw video frames and restricting outputs to anonymous density maps. Such maps

convey summary spatial information and are not designed to identify individuals, track trajectories, or support behavioural profiling. Nevertheless, even aggregate density data may, in some settings, be combined with other sources to infer information about individuals or communities. Therefore, technical safeguards should be complemented by institutional policies governing data retention, access, and permitted uses of system outputs.

In addition to privacy concerns, issues of bias and fairness must also be considered. Deep learning models trained on publicly available crowd-counting datasets may inherit biases arising from demographic diversity, geographical distribution, and environmental conditions represented in the data. Existing benchmark datasets such as NWPU-Crowd and ShanghaiTech are predominantly collected from urban scenes in East Asia [30] [15], which may limit model generalisation to other cultural or geographical contexts. As a result, density estimation accuracy may vary systematically across different crowd compositions, clothing styles, and environments. This thesis explicitly acknowledges this limitation, and future work should consider geographically and demographically diverse training and evaluation datasets.

Furthermore, automated crowd monitoring systems require clear mechanisms of accountability and transparency, particularly when they inform public safety decisions. Decision-makers should understand both the capabilities and limitations of the technology. The modular and documented architecture of the proposed system promotes transparency by specifying the role of each module and the conditions under which it is invoked. The open-source codebase and training pipeline further support external scrutiny and reproducibility [40]. However, the binary routing decision introduces an additional accountability concern. If the router incorrectly classifies a scene, the resulting density estimate may be significantly affected. Consequently, routing decisions and their limitations should be communicated to system operators.

Finally, the issue of informed consent becomes particularly relevant when monitoring occurs in public spaces. In many jurisdictions, passive observation of public areas is permitted provided that no personally identifiable information is recorded. Since the proposed system produces only non-identifying aggregate density maps, it generally falls within this category. Nevertheless, deployment decisions remain context-dependent, and operators are encouraged to consider relevant legal requirements and, where appropriate, communicate monitoring activities through signage or public disclosure.

3.5 Standards

The proposed system is designed and implemented according to a series of well known international standards and regulations. By complying with these standards, compatibility, interoperability, reproducibility and meeting the legal requirements are ensured.

The GDPR is a reference standard that is widely used throughout the world and serves as the legal framework for data protection in the European Union [32]. The edge-based, raw data-free design of the system follows the principles of GDPR, including data minimisation (only the minimum data needed is processed), purpose limitation (the data is processed only for the declared density estimation purpose), and storage limitation (raw frames are not stored). The density maps produced by the system are not processed as personal data under GDPR Article 4 and were

intentionally designed to avoid the processing of such data. In parallel, the National Institute of Standards and Technology Privacy Framework [41] provides a voluntary framework for privacy risk management. The proposed system aligns with its key functions: Identify (to understand the privacy risks associated with crowd monitoring), Govern (to establish operational policies that control data use), Control (to restrict processing to anonymised outputs), and Communicate (to maintain transparency with affected individuals where applicable).

Beyond privacy standards, the system also aligns with recognised ethical AI frameworks. The IEEE’s document on ethically aligned design for autonomous and intelligent systems [42] outlines principles of human wellbeing, transparency, accountability, and competence. The proposed system reflects these principles through privacy-preserving operation, open-source documentation, and a modular architecture that supports accountability and transparency. Furthermore, standards developed under the ISO/IEC JTC 1/SC 42 umbrella, including ISO/IEC TR 24028:2020, provide guidance on AI terminology, trustworthiness, robustness, and bias. These standards serve as a reference for understanding the societal implications of AI systems and inform the evaluation methodology adopted in this research.

In addition to privacy and ethical considerations, the system follows established software engineering standards. The evaluation methodology is based on IEEE 829 principles through the documentation of test goals, test environments, and evaluation metrics (MAE and MSE), ensuring that the evaluation results are reproducible and interpretable. Similarly, non-functional aspects of the system, including maintainability, modularity, portability, and efficiency, are defined through the system requirements and aligned with the quality characteristics specified in ISO/IEC 25010, which provides a comprehensive model for software product quality.

3.6 Project Management Plan

The project was carried out by a team with an adapted division of labor based on personal skills. Main responsibilities were related to the acquisition and preprocessing of data sets, model implementation (LCDNet, MobileCount, CSRNet, routing classifier), development of training scripts, evaluation and benchmarking, and documentation and thesis writing. The code was reviewed and integration tested by all the team members. To coordinate these activities, weekly meetings were held regularly to assign work, track progress, discuss technical problems, and discuss interim deliverables. The project supervisor held review sessions during the project development cycle.

The project was developed using a phased and iterative approach consisting of four main phases. Phase 1 focused on foundation and data preparation, including development environment setup, dataset acquisition (NWPU-Crowd, ShanghaiTech), implementation of the preprocessing pipeline, and preliminary baseline experiments on the ShanghaiTech dataset using LCDNet. Phase 2 involved the independent training of the routing classifier (MobileNetV2), LCDNet on the NWPU-Crowd sparse subset, and MobileCount on the NWPU-Crowd dense subset using the standard training pipeline. Phase 3 focused on knowledge distillation and system integration through the implementation of the CSRNet-to-MobileCount knowledge distillation pipeline, integration of all three components into the hybrid inference system, and implementation of both hard-routing and soft-fusion inference modes. Finally, Phase 4

consisted of evaluation, analysis and documentation, including full evaluation on the NWPU-Crowd validation set and ShanghaiTech cross-dataset benchmarks, threshold sweep analysis, router calibration evaluation, failure case analysis, and thesis writing.

Version control, collaborative development, and issue tracking were carried out throughout the project using a Git repository hosted on GitHub. A well-documented directory structure was used to maintain all training scripts, configuration files, pre-processing routines, and evaluation scripts within the repository. Checkpoint files were stored separately and referenced in the repository documentation to guarantee full repeatability of results. This approach follows the principles of reproducible machine learning research [43].

The project timeline consisted of four phases spanning eighteen weeks. Phase 1 (Weeks 1–4) covered environment setup, dataset preparation, and baseline experiments. Phase 2 (Weeks 5–9) focused on routing classifier training, LCDNet fine-tuning, and MobileCount training. Phase 3 (Weeks 10–13) involved knowledge distillation, system integration, and development of the inference pipeline. Phase 4 (Weeks 14–18) concentrated on evaluation, analysis, documentation, and thesis writing.

The project was developed using the following tools and infrastructure:

- **Programming Language:** Python 3.11.9
- **Deep Learning Framework:** PyTorch 2.6.0 (CUDA 12.4), torchvision 0.21.0
- **Training Hardware:** NVIDIA GeForce RTX 3050 (8 GB GDDR6)
- **Target Deployment Platform:** NVIDIA Jetson Nano (intended; physical benchmarking is future work)
- **Version Control:** Git 2.51 / GitHub
- **Documentation:** L^AT_EX
- **FLOPs Estimation:** thop (primary), fvcore (alternative)
- **Scientific Computing:** scipy 1.17.0 (used to .mat annotation parsing)

3.7 Risk Management

The proposed system is subject to a number of technical, ethical, and operational risks that have been identified and addressed through appropriate mitigation strategies. From a technical perspective, the routing classifier may incorrectly label a dense scene as sparse (or vice versa), directing the image to the less appropriate density estimator and degrading the overall count estimate. To mitigate this risk, the router is trained using a balanced dataset and evaluated using accuracy and Expected Calibration Error (ECE) metrics, while a configurable routing threshold (p^*) allows operators to tune the decision boundary according to deployment requirements [44]. In addition, both lightweight density estimators may undercount in extreme-density scenes because their architectural capacity is limited relative to heavier models. This limitation is partially addressed through the CSRNet knowledge distillation pipeline for MobileCount, while future work may explore multi-scale

crops or Bayesian loss formulations [45]. Another technical risk is dataset domain mismatch, where models trained on NWPU-Crowd may underperform in different geographic, environmental, or demographic contexts. Cross-dataset evaluation on ShanghaiTech is therefore included as a standard component of the evaluation plan, and domain adaptation techniques are identified as future work [46]. Compatibility and performance issues on specific edge hardware configurations also remain a possibility; however, the system targets POSIX-compliant Linux platforms supported by PyTorch, with ONNX export available as a fallback option. Furthermore, knowledge distillation training may diverge or converge to a suboptimal solution if loss weights or learning rate schedules are poorly configured. This risk is mitigated through a weighted combination of density supervision, teacher-to-student map matching, count-level L1 regularisation, cosine annealing, and gradient clipping [47]. Beyond technical considerations, ethical and regulatory risks must also be considered. Although density map outputs are non-identifying, they could potentially be combined with other data sources in ways that enable indirect identification or profiling. Consequently, deployment guidelines recommend strict access control policies and prohibit the combination of outputs with datasets that may enable re-identification. The system also avoids trajectory tracking and temporal linking capabilities. In addition, the system may exhibit differential accuracy across crowd demographics or geographic contexts not represented in the training data. This limitation is transparently disclosed in the system documentation, while evaluation across NWPU-Crowd and ShanghaiTech provides partial evidence of generalisation. Expanding training data diversity remains an important area for future work. Operational risks are comparatively limited. Access to target edge hardware such as the Jetson Nano may be delayed or restricted, potentially affecting deployment experiments. To reduce this dependency, core experiments are conducted on an RTX 3050 development platform, while edge deployment is treated as a validation stage and the software remains capable of CPU execution. Finally, access to benchmark datasets may be subject to licensing or usage restrictions. This risk is mitigated through the use of publicly available research datasets, including NWPU-Crowd and ShanghaiTech, whose documented terms permit academic research. In addition, preprocessing scripts generate derived density maps without redistributing the original raw data.

3.8 Economic Analysis

The development of the proposed hybrid crowd density estimation system was carried out within a university research environment, and the costs were primarily associated with researchers’ time and computing resources. All training and evaluation of the models were performed locally using the principal development hardware, an NVIDIA GeForce RTX 3050 GPU, without incurring any cloud computing expenses. The deployment hardware costs are also relatively low. Although commercial deployment was not undertaken as part of this research, the NVIDIA Jetson Nano, which serves as the intended deployment platform, is available at an approximate cost of USD 99–149 per unit [33]. Including peripheral components such as an enclosure, power supply, and camera module, the total per-node bill of materials remains below USD 200. This makes large-scale deployment economically feasible for organisations with limited budgets and compares favourably with cloud-based solutions,

where subscription fees typically increase with the number of inferences performed. The economic advantages of the edge-based design become even more apparent at scale. Organisations that transmit video streams to cloud inference services must bear both data transmission costs and per-inference API charges. These costs accumulate rapidly when multiple cameras operate continuously at high frame rates. In addition, cloud-based systems require reliable network connectivity, introducing further operational costs and risks. By contrast, once an edge device has been purchased and deployed, the primary operating cost is power consumption, which is approximately 5–10 W for the NVIDIA Jetson Nano [33]. There are no recurring API fees, bandwidth charges, or dependencies on cloud availability. Consequently, the total cost of ownership is significantly lower than that of comparable cloud-based solutions, particularly in high-frequency or remote deployment scenarios.

Beyond direct cost savings, the deployment of an accurate crowd density estimation system can generate substantial economic and social value across multiple application domains. The economic and social consequences of crowd disasters can be severe [48], and early warning systems help reduce the likelihood of such incidents. Accurate, real-time crowd distribution data also supports staffing decisions, resource allocation, and venue management, thereby reducing operational costs and improving visitor experiences. In urban planning, long-term crowd density information can contribute to more efficient infrastructure design, leading to substantial savings in transportation, security, and emergency services. Similarly, in healthcare and educational environments, occupancy monitoring can reduce manual counting tasks and support automatic compliance with safety protocols, saving valuable staff time. The long-term scalability of any system ultimately depends on its commercial viability, making economic sustainability an important consideration in practical deployment. The modular design of the proposed system further supports scalability and commercial adoption. Multiple edge devices can be deployed across different locations without requiring modifications to the core software pipeline, while system updates can be performed by replacing individual model checkpoints rather than redeploying the entire system. Such modularity [49], combined with the open-source codebase, facilitates third-party adoption and creates opportunities for integration into existing smart city and event management platforms. Furthermore, knowledge distillation improves the accuracy of the dense-scene specialist without increasing deployment costs, creating a commercially attractive development pattern in which performance gains are achieved during training through a larger teacher model, while deployment relies only on the lightweight student model. This approach can be extended in future iterations, where stronger teacher models may further enhance student performance without increasing the deployment footprint.

Chapter 4

Proposed Methodology

4.1 Design Process and Methodology Overview

The hybrid crowd density estimation system was designed by going through five stages: each stage was designed to solve a problem associated with a research objective and developed step by step based on the results of the previous stage. This systematic approach provides for an empirical approach to decision making, with the research objectives of each phase being tested against previous design decisions. The whole process of design is demonstrated in Figure 4.1.

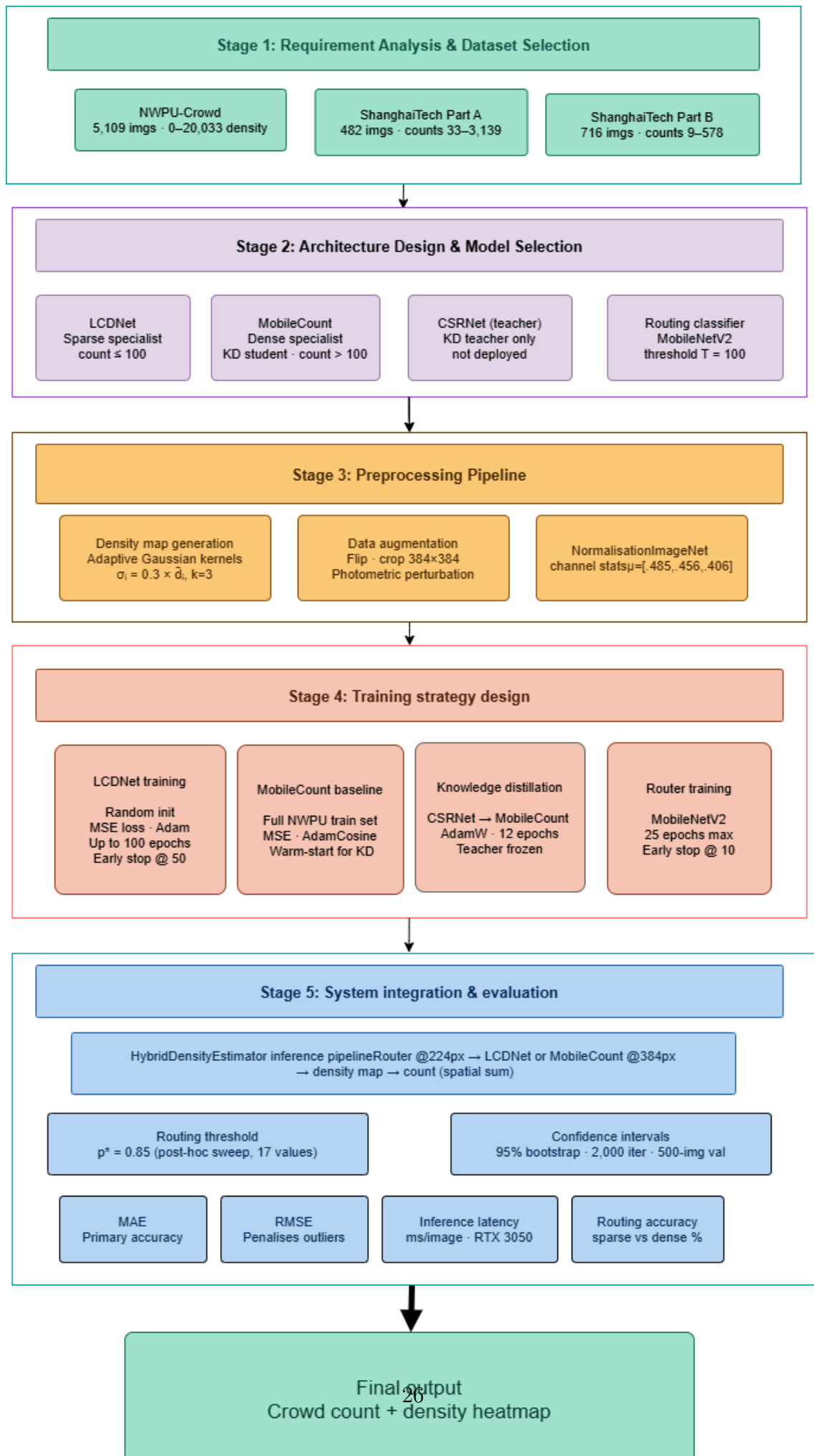
4.1.1 Phase 1: Requirement Analysis and Dataset Selection

The study methodology was focused on a preliminary stage, which involved gathering a comprehensive knowledge of the problem space and determination of suitable datasets for training and evaluating the proposed system. The selection was based on a set of criteria developed to ensure that the datasets selected represent the diversity and complexity of real-world crowd monitoring scenarios.

A number of system requirements were determined based on the deployment requirements and the problem area. The proposed system must be able to provide accurate density estimates in a variety of scenarios, such as low density (less than 100 people) and high density (more than 5,000 people). In addition to accuracy, the system needs to be compute efficient and capable of inference at a compute speed that is similar to the NVIDIA Jetson Nano. To maintain privacy, all video frame processing needs to be done on the device and cannot be sent to central servers as raw video frames. Moreover, the architecture should be modular and scalable, allowing the deployment of the network of cameras without much computational burden.

The choice of datasets was determined by the need that they would together cover the entire spectrum of crowd densities observed in the real world. The following criteria were followed: The dataset should contain diversity in the number of people to ensure coverage of both sparse and dense scenes, and point level annotations should be of adequate quality that allows for the creation of density maps, the dataset should contain enough training samples to ensure the convergence of the model, and it should be publicly available to ensure the ability to reproduce and compare with published methods.

Selected Datasets:



- **NWPU-Crowd [30]:** 5,109 images, 2,133,375 head annotations, crowd counts ranging from 0 to 20,033 per image. Chosen as the main training and evaluation indicator for the three implemented model components. Different scenes, lighting and camera angles.
- **ShanghaiTech Part A [15]:** 482 images (300 train, 182 test), 241,677 annotations, crowd counts ranging from 33 to 3,139. In this work, only used for a generalisation evaluation across different sets of data.
- **ShanghaiTech Part B [15]:** 716 images (400 train, 316 test), 88,488 annotations, crowd counts ranging from 9 to 578. Also for use in cross-dataset generalisation evaluation.

Note that all samples in ShanghaiTech Part A and Part B are only for evaluating zero-shot generalisation and they are not used for training, validation or hyperparameter tuning.

4.1.2 Phase 2: Architecture Design and Model Selection

The architecture design phase, after requirements analysis, focused on the trade-off between accuracy and efficiency in crowd density estimation and a new hybrid architecture was developed that can adaptively select a model according to the level of complexity of the input scene.

Architectural Reasoning:

The main problem with the estimation of crowd density is that it is important to provide accurate estimation while keeping the computation time as low as possible. Lightweight models are fast for inference but have low representation ability for dense scenes. Highly accurate models require a lot of computational resources, which is not suitable for their use at the edges and is an unnecessary overhead in dense inputs. One way to resolve this tension is to use a conventional approach of choosing a single model and operation, which results in sub-optimal performance throughout the density spectrum. The proposed hybrid architecture addresses this drawback by dynamically switching between two specialised lightweight models: one to process sparse scenes, and one to process dense scenes, as per a learned evaluation of the level of complexity of the scene.

Architectural Components:

Component 1: LCDNet (Lightweight Crowd Density Network). Intended to be applied on scenes with small to medium object density (ground truth object count ≤ 100). The architecture is based on an encoder-decoder design using depthwise separable convolutions which has about 8 times fewer parameters than traditional convolutional architectures of the same depth but have the same representational capacity. Skip connections connect stages in both the encoder and decoder, maintaining the spatial information at a fine level of granularity through the downsampling hierarchy. LCDNet creates full resolution density maps of the same spatial dimensions as the input.

Component 2: MobileCount (Deployed Dense Specialist; Knowledge-Distilled).

For high-density crowd situations (ground-truth number > 100). Uses the first 13 feature layers of a pretrained MobileNetV2 backbone as the encoder and

uses the feature layers to extract features at multiple levels to provide hierarchical feature extraction that is computationally efficient and generalises well from the ImageNet pretraining. Three dilated convolution context blocks (Dilation Rate = 2) are used for multi-scale contextual information, without low resolution. A skip-connection fusion module fuses representations of the encoders at two scales of resolution for better density-map accuracy. The result is a density map of 1/8 resolution compared to the input map and the crowd count is obtained by spatial summation. MobileCount is deployed in the knowledge distilled form, and is trained to match the density map output of a CSRNet teacher. After distillation, the teacher is thrown away and is not part of the pipeline deployed.

Component 3: CSRNet (Knowledge Distillation Teacher; Not Deployed).

Only used as the teacher during the phase of knowledge distillation when it is used during the training of a teacher. Network structure: Pretrained VGG-16 convolutional layers (first 10) as the front end and a density estimator with six dilated convolutional layers (dilation rate: 2, 2, 2, 4, 4, 4). It has a large number of parameters (16.26M) and is not suitable for direct deployment on the edge; but its edge capability of dense-scene density estimation is distilled into the lightweight MobileCount student.

Component 4: MobileNetV2 Routing Classifier. An optimized MobileNetV2 backbone plus a customized two-class fully connected head network that classifies the input as sparse (count ≤ 100) or dense (count > 100). The routing threshold $T = 100$ is similar to previous research on crowd counting that classified the crowd as sparse or dense.

4.1.3 Phase 3: Preprocessing Pipeline

The phase achieves the final data preprocessing pipeline, generating and applying density maps, data augmenting and normalizing to create a set of standard training representations systematically in a multi-stage process by applying raw annotated imagery.

Stage 1: Density Map Generation.

Ground truth continuous density maps are generated from head point annotations by creating adaptive Gaussian kernels along each head point annotation. The average of the standard deviations σ_i of the kernels at the point of the annotation i is:

$$\sigma_i = \beta \cdot \bar{d}_i$$

The mean Euclidean distance between the position of the head and its $k = 3$ nearest neighbours within the same image, $\bar{d}_i$, is used. The scaling coefficient, $\beta = 0.3$, is determined empirically. This adaptive formulation results in narrower kernels in dense regions (maintaining localisation accuracy) and wider kernels in sparse regions (ensuring sufficient spatial coverage). Conservation of mass is respected all throughout: $\iint D(x, y) dx dy = N$, where N is the number of heads annotated in the image, which allows for ground-truth supervision by pixel-wise regression.

If an image has just one person annotated, and it's not possible to compute nearest-neighbour distance, a default sigma value of 15.0 pixels is used. To avoid producing

numerically unstable training signals due to degenerate kernels all computed sigma values are clipped to the range $[4, 30]$ pixels.

Stage 2: Data Augmentation.

The following Stochastic Augmentation Pipeline is used during training to increase the generalisation of the model and the reduction of overfitting. To ensure the pixel-level correspondence of the image and the density map, all geometric transformations are applied to the image and its density map equally:

1. Random horizontal flip of 50% probability.
2. Random 384×384 patch crop from a 512×512 rescaled image (probability 0.3).
3. Photometric perturbation: brightness $\pm 20\%$, contrast $\pm 20\%$, saturation $\pm 20\%$ and hue perturbation.

Stage 3: Normalisation.

All images are normalised using ImageNet channel statistics (mean: $[0.485, 0.456, 0.406]$; standard deviation: $[0.229, 0.224, 0.225]$ for R, G, B channels respectively). It is a normalisation scheme that ensures any distribution of input matches the distributions of pre-trained weights of the MobileNetV2 backbones in both MobileCount and the routing classifier, which helps to achieve fast convergence and good feature transfer.

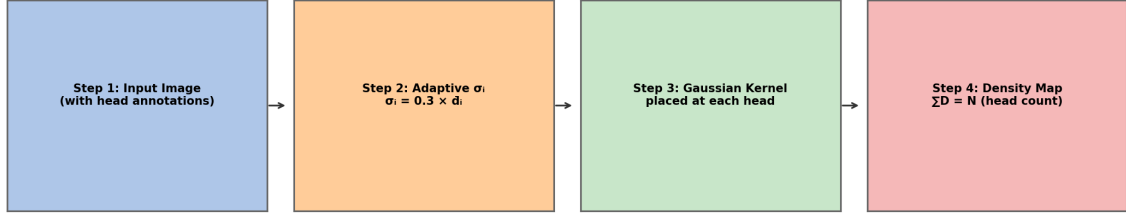


Figure 4.2 — Adaptive Gaussian Density Map Generation from Point Annotations

Figure 4.2: Adaptive Gaussian Density Map Generation from Point Annotations. Four-stage pipeline: (1) raw crowd image with head annotations, (2) computation of adaptive kernel width for each head, $\sigma_i = 0.3 \times \bar{d}_i$ ($k = 3$ nearest neighbours), (3) Gaussian kernel placement at each head, (4) final continuous density heatmap with $\sum D = N$ enforced.

4.1.4 Phase 4: Training Strategy Design

The training strategy varies for each model component, depending on the architecture features and the part they play in the deployed hybrid system. In particular, the following is a description of the training procedures for the three components.

LCDNet Training.

The initial weights of LCDNet are trained randomly on the sparse training subset of NWPU-Crowd (images with the ground-truth count < 100). The goal of the training is to minimize the pixel-by-pixel mean squared error (MSE) between the density map predictions and ground truth. An Adam optimiser with learning rate 1×10^{-4} and weight decay 1×10^{-5} are used. ReduceLROnPlateau learning rate

scheduler decreases the learning rate by 0.5 if the validation loss doesn't improve for 15 epochs. The model was trained for as many as 100 epochs with early stopping after 50 epochs. Gradient clipping is used to avoid training instability to a maximum norm of 5.0.

MobileCount Baseline Training.

To train MobileCount, it is trained using the full training set of NWPU-Crowd, supervised by MSE on 1/8-resolution density maps. The Adam optimiser with a learning rate of 1×10^{-4} and cosine annealing learning rate schedule is used. The warm start initialisation for the knowledge distillation phase is this baseline checkpoint.

Knowledge Distillation; CSRNet Teacher to MobileCount Student.

Subsequently, MobileCount is further improved by knowledge distillation from the teacher CSRNet (frozen) model. The loss during the combined distillation is:

$$L_{distill} = \alpha \cdot \text{MSE}(D_{student}, D_{GT}) + \beta \cdot \text{MSE}(D_{student}, D_{teacher}) + \gamma \cdot L_1(C_{student}, C_{GT})$$

Here $D_{student}$ and $D_{teacher}$ are the density map outputs for the student and teacher, respectively; D_{GT} is the ground-truth density map; $C_{student}$ is the number of students the student predicted, or the spatial sum of $D_{student}$; C_{GT} is the number of students in the ground truth; and loss coefficients are $\alpha = 0.5$, $\beta = 0.5$, $\gamma = 0.05$. AdamW (learning rate = 5×10^{-5} and weight decay = 1×10^{-4}) is used to optimise the student, which is run for 12 epochs on a cosine annealing schedule ($\eta_{min} = 1 \times 10^{-7}$). Gradient Clipping to Max Norm 1.0. All of the teacher's parameters remain constant during the entire time (`requires_grad = False`). Finally, after the training, the checkpoint having the minimum validation MAE is kept and the teacher is deleted.

Routing Classifier Training.

Finally, the routing classifier is fine-tuned on a pretrained MobileNetV2 backbone from a binary cross-entropy loss. The classification head is a custom classification head that is randomly initialised and trained from scratch, $\text{Dropout}(p = 0.3) \rightarrow \text{Linear}(1280 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(p = 0.3) \rightarrow \text{Linear}(256 \rightarrow 2)$. An Adam optimiser is used (learning rate = 1×10^{-4} , weight decay = 1×10^{-5}). Training for up to 25 epochs (with EarlyStopping at epoch 10 with no significant gain in validation accuracy).

4.1.5 Phase 5: System Integration and Evaluation

The last step is to combine all three deployed components to a whole inference pipeline and compare the system with known benchmarks. Several performance measures are used to evaluate the performance of the framework.

Performance Measures

- **MAE:** The mean absolute error of the predictions and count data, which is the main accuracy metric.
- **RMSE:** Root mean squared error – penalises large individual errors more than MAE.
- **Inference Latency:** Per-image inference time (in ms) measured on the RTX 3050, after 5 warm-up passes, with 30 evaluation passes.

Figure 4.3 — Knowledge Distillation Architecture: CSRNet Teacher → MobileCount Student

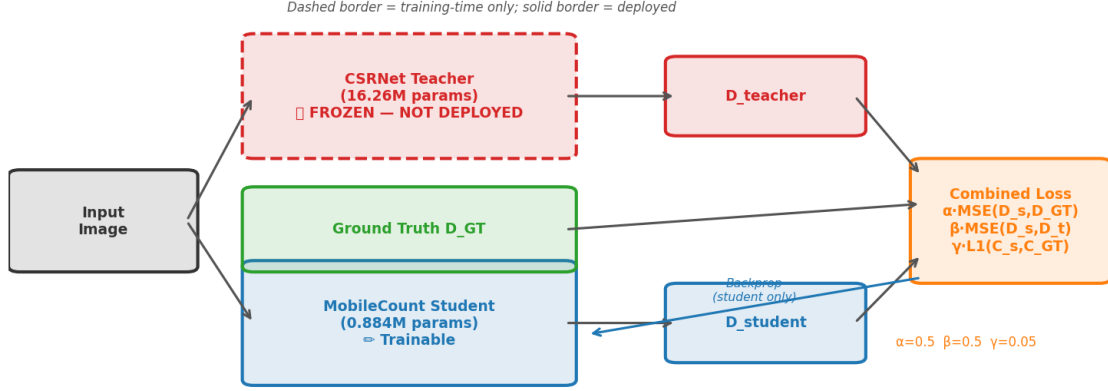


Figure 4.3: Knowledge Distillation Architecture. $D_{teacher}$ is the result of the frozen CSRNet teacher (16.26M params, dashed border, training only, not deployed). The trainable student MobileCount (0.884M, solid border) creates $D_{student}$ (solid border). Three loss terms: $\alpha \cdot \text{MSE}(D_{student}, D_{GT})$, $\beta \cdot \text{MSE}(D_{student}, D_{teacher})$, $\gamma \cdot L_1(C_{student}, C_{GT})$ are summed; the gradient is only calculated for the student. The teacher is completely removed from the deployed system after training.

- **Routing Accuracy:** Accuracy of the correct classification of the images (sparse vs dense) in the NWPU-Crowd val set.
- **95% Bootstrap Confidence Intervals:** Based on 2,000 iteration bootstrap resampling of the 500 image validation set.

Integration Architecture.

Once these evaluation metrics are defined, all three are combined in the `HybridDensityEstimator` class. The input images are all passed to the routing classifier (224×224) to generate class probabilities $[P(\text{sparse}), P(\text{dense})]$. If the $P(\text{dense}) \geq p^*$, the image is sent to MobileCount at 384×384 , else to LCDNet at 384×384 . The routing threshold p^* is optimized post hoc by sweeping 17 possible values of p^* on the validation set, and determining that $p^* = 0.85$ is the optimal operating point which provides the minimum system MAE without retraining the model.

4.2 Preliminary Design and Model Specification

4.2.1 Dataset Characteristics

First of all, when designing a dataset for crowd density estimation, one needs to pay attention to the annotation quality, variety of scenes, the differences of density and the applicability of deployment scenarios in reality. This work uses two complementary benchmark datasets: NWPU-Crowd as the main training and evaluation benchmark; and ShanghaiTech as the cross-dataset generalisation evaluation

benchmark. These data sets are briefly summarised in Table 4.1 in terms of their statistical properties.

Table 4.1: Dataset Statistical Characteristics

Characteristic	ShanghaiTech Part A	ShanghaiTech Part B	NWPU-Crowd
Total Images	482	716	5,109
Total Annotations	241,677	88,488	2,133,375
Training Images	300	400	3,109
Validation Images	—	—	500
Test Images	182	316	1,500
Min Crowd Count	33	9	0
Max Crowd Count	3,139	578	20,033
Mean Count	501.4	123.6	417.5
Standard Deviation	389.2	90.4	791.4
Image Resolution	Variable ($\sim 768 \times 1024$)	Variable ($\sim 768 \times 1024$)	Variable ($\sim 2,191 \times 3,209$)
Usage in this work	Cross-dataset eval only	Cross-dataset eval only	Primary (all components)

Moreover, NWPU-Crowd has a large distribution of density ranging from totally empty frames to a maximum of more than 20,000 people in a single frame, which is well-suited for training all three parts of the system in the entire operating range. To support complementary evaluation of zero-shot generalisation, ShanghaiTech Part A offers dense-to-very-dense scenes and Part B offers mostly sparse street-level scenes.

4.2.2 LCDNet Architecture Specification

Design Philosophy:

By explicitly targeting the efficiency of parameters, LCDNet adopts depthwise separable convolutions, which lead to about 1/8 the parameters of standard convolutions of the same representational depth, and preserves enough capacity to correctly estimate the density of the crowd in the images, which are sparse to medium. The design will be used as the "fast path" for the hybrid system when the scene has less than 100 people.

Architecture Overview:

LCDNet uses a symmetric skip connections encoder-decoder architecture with a fine-grained spatial information through the downsampling hierarchy. The encoder performs four successive downsamplings, each increase the channel depth. Each encoder block is made up of two depthwise separable convolution (DSConv) layers (each layer is a depthwise convolution followed by a pointwise convolution (1×1), batch normalisation, and ReLU activation) and then followed by MaxPool 2×2 . The bottleneck (Stage 4) does not include the pooling. Each decoder stage upsamples (bilinear) the features, concatenates the upsampled features with the corresponding encoder skip connection and applies two DSConv layers to further refine the features. The output head takes the last 64-channel feature map and transforms it to a density map, in such a way that the density is always non-negative, by passing it through Conv $3 \times 3(64 \rightarrow 32) + \text{ReLU} + \text{Conv}1 \times 1(32 \rightarrow 1) + \text{ReLU}$. LCDNet adopts this design strategy, and uses a symmetric skip-connections architecture with an encoder-decoder to maintain fine-grained level-by-level spatial information in the downsampling hierarchy. The detailed layer wise specification is given in Table 4.2.

Table 4.2: LCDNet Detailed Layer-wise Specification

Component	Layer Configuration	Dimensionality Transform	Parameters	Functional Role
Encoder Stage 1	DSCnv $\times 2$ + MaxPool2 $\times 2$	$384 \times 384 \times 3 \rightarrow 192 \times 192 \times 64$	$\sim 3,500$	Primary feature extraction
Encoder Stage 2	DSCnv $\times 2$ + MaxPool2 $\times 2$	$192 \times 192 \times 64 \rightarrow 96 \times 96 \times 128$	$\sim 17,000$	Intermediate representations
Encoder Stage 3	DSCnv $\times 2$ + MaxPool2 $\times 2$	$96 \times 96 \times 128 \rightarrow 48 \times 48 \times 256$	$\sim 66,000$	Mid-level semantic features
Encoder Stage 4 (Bottleneck)	DSCnv $\times 2$ (no pool)	$48 \times 48 \times 256 \rightarrow 48 \times 48 \times 512$	$\sim 264,000$	High-level abstractions
Dropout2D	Regularisation	$48 \times 48 \times 512$	0	Prevent overfitting (p=0.2)
Decoder Stage 1	Upsample + Concat + DSCnv $\times 2$	$48 \times 48 \times 512 \rightarrow 96 \times 96 \times 256$	$\sim 396,000$	Initial spatial reconstruction
Decoder Stage 2	Upsample + Concat + DSCnv $\times 2$	$96 \times 96 \times 256 \rightarrow 192 \times 192 \times 128$	$\sim 99,000$	Intermediate upsampling
Decoder Stage 3	Upsample + Concat + DSCnv $\times 2$	$192 \times 192 \times 128 \rightarrow 384 \times 384 \times 64$	$\sim 25,000$	Fine-grained reconstruction
Output Head	Conv3 $\times 3$ (64 $\rightarrow$ 32)+ReLU+Conv1 $\times 1$ (32 $\rightarrow$ 1)+ReLU	$384 \times 384 \times 64 \rightarrow 384 \times 384 \times 1$	$\sim 18,500$	Density map generation
Total	—	—	$\approx$ 0.917M	Full-resolution output

DSCnv = Depthwise Conv + Pointwise Conv + BN + ReLU. Skip connections are used to connect pre-pool encoder features with upsampled decoder features at every resolution.

Figure 4.4 — LCDNet Encoder-Decoder Architecture with Skip Connections

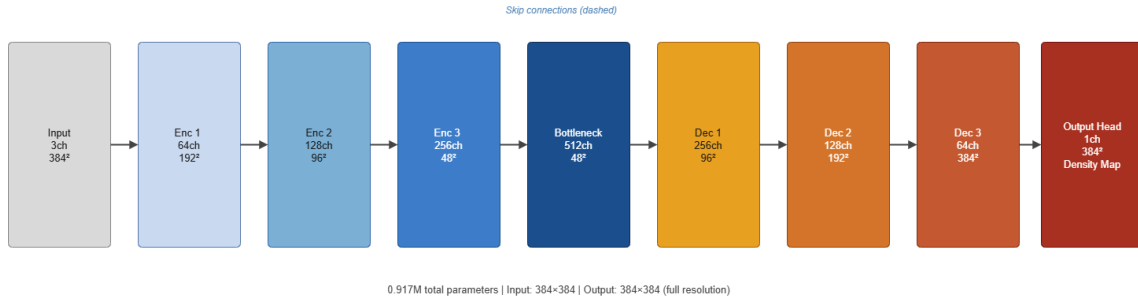


Figure 4.4: LCDNet Encoder-Decoder Architecture with Skip connections. The depth of each channel and spatial resolution for each are labeled. Skip connections are identified in dashed arrows, from encoder to decoder. Parameters used: 0.917M; output resolution is same as input (384 $\times$ 384).

4.2.3 MobileCount Architecture Specification (Dense Specialist)

Design Philosophy:

The high capacity path of the hybrid system, on the other hand, is to be used in dense crowd scenes, where MobileCount is designed to work. The architecture benefits from the speed and versatility of a pre-trained MobileNetV2 encoder, while adding dilated convolution (DC) blocks for context and multi-scale feature extraction, and a skip-connection fusion module that recovers spatial details.

Architecture Overview:

The first 13 feature layers in the MobileNetV2 model are split into two stages, and implemented as the Encoder. As a result, the encoder is split into 2 stages in the first 13 layers of MobileNetV2 feature layers. The spatial feature map, which is the result of Stage 1 (`backbone.features[0:7]`), is stored as a skip connection, and is a 1/8-resolution feature map of the same 32 channels produced by this stage. Stage 2 (`backbone.features[7:14]`) generates a 1/16-resolution, 96 channel image. Multi-scale contextual information is extracted by three dilated convolution blocks ($\text{Conv3} \times 3$, dilation=2, $96 \rightarrow 96$ channels). Features are upsampled by $\times 2$ to match the resolution of Stage 1, and then concatenated with the skip connection ($32\text{ch} + 96\text{ch} = 128\text{ch}$). A fusion block ($\text{Conv3} \times 3$ $128 \rightarrow 64$ + BN + ReLU + $\text{Conv3} \times 3$ $64 \rightarrow 32$ + BN + ReLU) integrates the combined features. The output head ($\text{Conv1} \times 1$ $32 \rightarrow 1$ + ReLU) produces the final 1/8-resolution density map (48×48 for 384×384 input). Table specifications are included in their entirety in Tables 4.3 and 4.4.

Table 4.3: MobileCount Architecture Specification

Component	Configuration	Output Shape	Notes
Input	RGB image	$B \times 3 \times 384 \times 384$	Standard resolution
Encoder Stage 1 (<code>features[0:7]</code>)	MobileNetV2 inverted residuals	$B \times 32 \times 48 \times 48$	Stride 8; skip connection source
Encoder Stage 2 (<code>features[7:14]</code>)	MobileNetV2 inverted residuals	$B \times 96 \times 24 \times 24$	Stride 16
Dilated Context Block 1	$\text{Conv3} \times 3$ ($96 \rightarrow 96$, dilation=2) + BN + ReLU	$B \times 96 \times 24 \times 24$	Multi-scale context
Dilated Context Block 2	$\text{Conv3} \times 3$ ($96 \rightarrow 96$, dilation=2) + BN + ReLU	$B \times 96 \times 24 \times 24$	Expanded receptive field
Dilated Context Block 3	$\text{Conv3} \times 3$ ($96 \rightarrow 96$, dilation=2) + BN + ReLU	$B \times 96 \times 24 \times 24$	—
Bilinear Upsample $\times 2$	—	$B \times 96 \times 48 \times 48$	Match Stage 1 resolution
Concat (skip + upsampled)	—	$B \times 128 \times 48 \times 48$	32ch + 96ch
Fusion Block	$\text{Conv3} \times 3$ ($128 \rightarrow 64$) + BN + ReLU + $\text{Conv3} \times 3$ ($64 \rightarrow 32$) + BN + ReLU	$B \times 32 \times 48 \times 48$	Feature integration
Output Head	$\text{Conv1} \times 1$ ($32 \rightarrow 1$) + ReLU	$B \times 1 \times 48 \times 48$	Density map (1/8 res)
Total Parameters	—	—	$\approx 0.884\text{M}$

Count = $\text{sum}(\text{density_map})$. Distilled in the knowledge to be deployed. Inference: 3.4 ms GPU, 13.5 ms CPU.

Table 4.4: CSRNet Teacher Architecture Reference (Training Only ; Not Deployed)

Component	Layer Configuration	Dimensionality Transform	Parameters	Status
Frontend VGG Block 1-2	$2 \times \text{Conv3-64}$, Pool, $2 \times \text{Conv3-128}$, Pool	$384 \times 384 \times 3 \rightarrow 96 \times 96 \times 128$	260,160	ImageNet pretrained
Frontend VGG Block 3	$3 \times \text{Conv3-256}$, Pool	$96 \times 96 \times 128 \rightarrow 48 \times 48 \times 256$	1,475,328	ImageNet pretrained
Frontend VGG Block 4-5	$6 \times \text{Conv3-512}$	$48 \times 48 \times 256 \rightarrow 48 \times 48 \times 512$	12,979,200	ImageNet pretrained
Backend Dilated 1-3	$3 \times \text{Conv3-512}$ (dilation=2)	$48 \times 48 \times 512 \rightarrow 48 \times 48 \times 512$	7,079,424	Randomly initialised
Backend Dilated 4-6	Conv3-256 , Conv3-128 , Conv3-64 (dilation=4)	$48 \times 48 \times 512 \rightarrow 48 \times 48 \times 64$	1,548,736	Randomly initialised
Output	1×1 Conv + ReLU	$48 \times 48 \times 64 \rightarrow 48 \times 48 \times 1$	65	Randomly initialised
Total	—	—	16,263,489	Not deployed; KD teacher only

4.2.4 Routing Classifier Specification

Design Philosophy:

The routing classifier is designed to be quickly and accurately trained for binary

labelling of scene complexity, and works well without requiring a significant amount of computational power, by using transfer learning on the visual representations of the backbone network (MobileNetV2), which is trained on the pretrained model.

Architecture:

In this way, the routing classifier is configured with the MobileNetV2 (pretrained on ImageNet) backbone network instead of the network’s classification head, and there is a two-class classification head for it. After the backbone’s adaptive average pooling layer produces a 1280-dimensional feature vector, the classification head is:

$$\text{Dropout}(p = 0.3) \rightarrow \text{Linear}(1280 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(p = 0.3) \rightarrow \text{Linear}(256 \rightarrow 2)$$

Logits for 2-class is transformed to the probability (p) of the class being sparse or dense using softmax: $[P(\text{sparse}), P(\text{dense})]$. Action taken in the routing decision: If $P(\text{dense}) \geq p^*$, send to MobileCount, if $P(\text{dense}) < p^*$, send to LCDNet.

Table 4.5: Routing Classifier Architecture Specification

Component	Type	Dimensionality	Parameters	Training Status
MobileNetV2 Backbone	Feature extractor	$224 \times 224 \times 3 \rightarrow 1280$	2,257,984	Fine-tuned (pretrained init)
Global Average Pooling	Spatial reduction	1280	0	—
Dropout Layer 1	Regularisation	1280	0	$p = 0.3$
Linear Layer 1	Fully connected	$1280 \rightarrow 256$	327,936	Trainable
ReLU	Activation	256	0	—
Dropout Layer 2	Regularisation	256	0	$p = 0.3$
Linear Layer 2 (Head)	Fully connected	$256 \rightarrow 2$	514	Trainable
Total	—	—	$\sim 2.552\text{M}$	—

Routing label: sparse (0) if GT count ≤ 100 ; dense (1) if GT count > 100 . Threshold $T = 100$.

4.2.5 Preprocessing Specifications

Density Map Synthesis:

Ground-truth density maps are generated by adding adaptive Gaussian kernels to the locations of the heads annotated. The standard deviation at the i -th head position σ_i of the kernel is:

$$\sigma_i = \beta \cdot \bar{d}_i$$

where $\bar{d}_i = \frac{1}{3} \sum_{j=1}^3 \|p_i - p_j\|_2$ is the mean Euclidean distance from annotation i to its three closest neighbours ($k = 3$), and $\beta = 0.3$ is an empirically determined scaling coefficient.

This gives rise to a density map $D(x, y)$ with the property that:

$$\iint D(x, y) dx dy = N$$

where N is the number of annotated heads in the image, allowing to regress each pixel in a supervised learning framework.

4.3 Selected Design Implementation

4.3.1 Implementation Environment

The Implementation Environment is summarised according to the model design specification as shown in Tables 4.6 and 4.7.

Table 4.6: Hardware Infrastructure Specification

Component	Technical Specification	Role
GPU	NVIDIA GeForce RTX 3050 (8 GB GDDR6 VRAM)	Model training and inference
CUDA	12.4 with cuDNN	GPU acceleration for operations
System Memory	16 GB DDR4	Dataset caching and processing
Operating System	Windows 11 Pro (64-bit)	Host environment

Note: It is designed for use at the edge level with the NVIDIA Jetson Nano. Future work identified for physical jetson is physical jetson benchmarking. In this thesis all the latency measurements are taken on the RTX 3050 (GPU) and single thread CPU.

Table 4.7: Software Ecosystem Versions

Package	Version	Role
Python	3.x	Primary runtime environment
PyTorch	2.6.0+cu124	Deep learning framework
torchvision	0.21.0+cu124	Vision models and transforms
NumPy	Latest compatible	Numerical computation
SciPy	1.17.0	Scientific computing, .mat file parsing
thop	Latest	FLOPs measurement
fvcore	Latest	FLOPs measurement (alternative)

4.3.2 Training Methodology and Configuration

4.3.3 Experimental Findings and Performance Analysis

LCDNet Evaluation

LCDNet was first tested on the entire 500-image NWPU-Crowd validation set. The full-set MAE value of 348.0 is for the exposure of the model to the whole range of density, including dense and extreme density scenes, which are not in the specialised regime for which the model was intended. For the sparse stratum, where the ground-truth number of items is ≤ 100 ($n = 181$), LCDNet achieves an MAE of 21.5, showing good performance in the specified stratum. The performance metrics are presented in Table 4.12.

MobileCount Knowledge Distillation Evaluation

Then, MobileCount was tested using the entire 500-image NWPU-Crowd validation set both in the original and teacher-improved versions. The performance of both configurations is summarised in Table 4.13 and the per epoch progression of the distillation training is recorded in Table 4.14.

Table 4.8: LCDNet Training Hyperparameters

Hyperparameter	Value	Justification
Dataset	NWPU-Crowd sparse subset (count ≤ 100)	Domain specialisation for s
Batch Size	8	Within GPU VRAM budget
Max Epochs	100 (early stopping: patience 50)	Allow full convergence
Optimiser	Adam	Adaptive learning rate
Learning Rate	1×10^{-4}	Standard for Adam optimiser
LR Schedule	ReduceLROnPlateau (factor=0.5, patience=15)	Adaptive reduction on plateau
Loss Function	MSE (density map)	Pixel-wise regression
Weight Decay	1×10^{-5}	L2 regularisation
Gradient Clipping	5.0 (max norm)	Training stability
Augmentation	Horizontal flip, random crop, colour jitter	Generalisation

Table 4.9: MobileCount Baseline Training Hyperparameters

Hyperparameter	Value
Dataset	NWPU-Crowd full training set (3,109 images)
Batch Size	8
Max Epochs	50
Optimiser	Adam (lr = 1×10^{-4})
LR Schedule	Cosine Annealing
Loss Function	MSE (density map at 1/8 resolution)

Table 4.10: Knowledge Distillation Configuration

Hyperparameter	Value	Notes
Student initialisation	Baseline MobileCount checkpoint	Warm start
Teacher	CSRNet (16.26M, all params frozen)	<code>requires_grad = False</code>
Epochs	12	Best checkpoint at epoch 9
Batch Size	8	—
Optimiser	AdamW	Decoupled weight decay
Learning Rate	5×10^{-5}	Fine-tuning rate
Weight Decay	1×10^{-4}	Stronger regularisation during distillation
LR Schedule	Cosine Annealing ($\eta_{min} = 1 \times 10^{-7}$)	Smooth decay
Gradient Clipping	1.0 (max norm)	Training stability
α (GT density weight)	0.5	Ground-truth supervision
β (teacher KD weight)	0.5	Teacher imitation term
γ (count L1 weight)	0.05	Count-level regularisation
Save path	<code>checkpoints/mobilecount_distilled.pth</code>	—

Figure 4.5: LCDNet Training and Validation Loss Curves. Plot of training loss (blue, solid) and validation loss (orange, dashed) against epochs. Reduction events of the learning rate are indicated by vertical dashed lines. This paper is entitled "LCDNet Training Convergence ; NWPU-Crowd Sparse Subset.

Table 4.11: Routing Classifier Training Hyperparameters

Hyperparameter	Value	Justification
Dataset	NWPU-Crowd (3,109 train, 500 val)	Binary labels from GT count
Batch Size	32	Classification task
Max Epochs	25 (early stopping patience = 10)	Rapid convergence
Optimiser	Adam	Fine-tuning
Learning Rate	1×10^{-4}	Standard
Weight Decay	1×10^{-5}	L2 regularisation
Loss Function	Binary Cross-Entropy	Binary classification
Dropout	$p = 0.3$ (both head layers)	Prevent overfitting
Best checkpoint	Epoch 10 (Val Accuracy: 88.80%)	Highest validation accuracy
Total training time	85.6 minutes (RTX 3050)	—

Table 4.12: LCDNet Performance Metrics ; NWPU-Crowd Validation Set

Metric	Full Val Set ($n = 500$)	Sparse Stratum Only ($n = 18$)
MAE	348.0	21.5
RMSE	1,033.9	—
Median Absolute Error	98.1	—
95% CI (MAE)	[269.5, 443.1]	—
Inference Time (GPU, RTX 3050)	23.1 ms	—
Inference Time (CPU, single-thread)	208.8 ms	—
GPU Peak Memory	368.8 MB	—
GFLOPs	14.59	—
Parameters	0.917M	—

The full set MAE is high, as LCDNet is the occasional specialist, and is called only for images in the hybrid system of count ≤ 100 .

Figure 4.6: LCDNet Qualitative Predictions. Set up 4-6 predictions of what will happen in each scene. Each column contains: (a) input image, (b) ground truth density map, (c) predicted density map and (d) absolute error map. Samples taken from areas with low density (GT count 100 or less).

Table 4.13: MobileCount Performance Metrics ; NWPU-Crowd Validation Set

Metric	Baseline (no KD)	Distilled (KD from CSRNet)	Improvement
MAE	213.2	198.7	−14.5
RMSE	803.6	778.5	−25.1
Median Absolute Error	—	58.7	—
95% CI (MAE)	[155.4, 292.6]	[143.8, 275.1]	—
Inference Time (GPU)	3.4 ms	3.4 ms	0
Inference Time (CPU)	13.5 ms	13.5 ms	0
GPU Peak Memory	63.6 MB	63.6 MB	0
GFLOPs	1.07	1.07	0
Parameters	0.884M	0.884M	0

Knowledge distillation, with zero deployment cost, same parameters as baseline, GFLOPs, and latency, contributes 14.5 MAE points to the accuracy of the standalone.

Table 4.14: Knowledge Distillation Per-Epoch Validation MAE

Epoch	Validation MAE	Notes
0	239.6	Initial post-distillation loss shift from baseline
1	217.6	Rapid initial improvement
2	216.2	—
3	208.9	Stable improvement phase
4	227.9	Temporary increase
5	200.2	Crosses below baseline MAE (213.2)
6	203.8	—
7	204.4	—
8	215.6	—
9	198.7	Best checkpoint ; saved
10	198.8	Marginal degradation
11	200.2	Training complete
Baseline (pre-distillation)	213.2	Reference

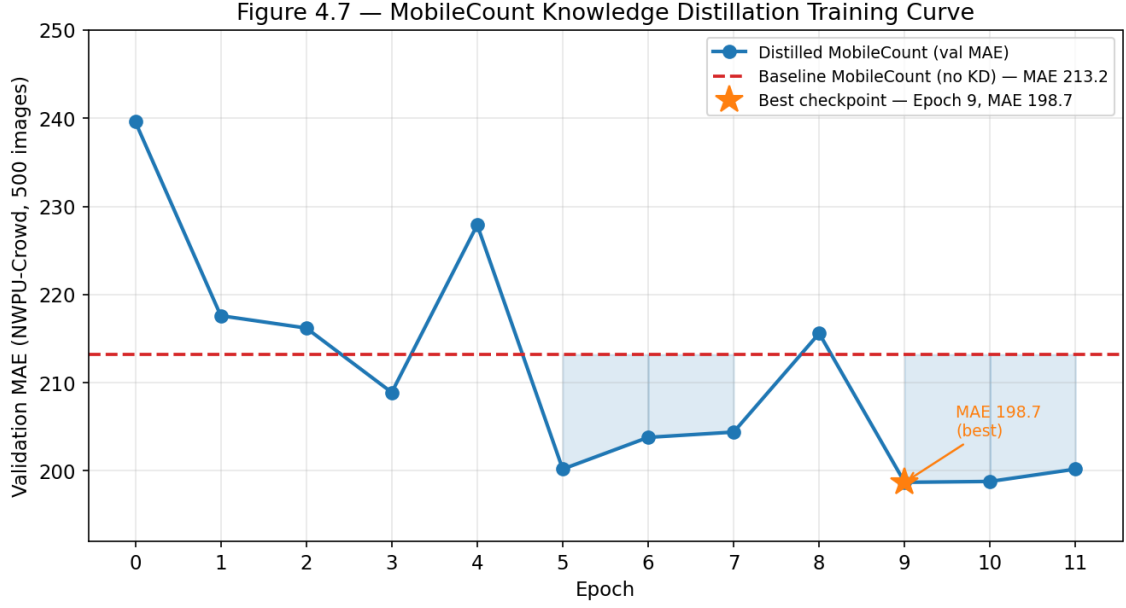


Figure 4.7: MobileCount Knowledge Distillation Training Curve. Validation MAE per epoch (blue line with markers, epoch 0-11) versus baseline MAE (red dashed line, MAE 213.2) before distillation. The best checkpoint at epoch 9 (MAE 198.7) is marked by the gold star. The epochs highlighted in blue represent periods when the distilled model is more accurate than the baseline.

Routing Classifier Evaluation

Afterwards, the effectiveness of the routing classifier in adaptive model selection was evaluated. Table 4.15 lists the training dynamics of the model at some selected epochs and Table 4.16 shows the final performance of the model. The confusion matrix and precision/recall breakdown show the classification behaviour at the argmax routing threshold.

Table 4.15: Routing Classifier Training Dynamics (Selected Epochs)

Epoch	Train Loss	Train Acc (%)	Val Loss	Val Acc (%)	Notes
1	0.4086	80.48	0.2887	87.60	Rapid initial learning
5	0.0944	96.62	0.3810	87.00	Training accuracy improving
10	0.0125	99.61	0.5096	88.80	Best validation accuracy
15	0.0114	99.65	0.6424	87.40	Validation plateauing
20	0.0072	99.71	0.5792	86.80	Early stopping triggered

The training was stopped at epoch 20 (early stopping 10 epochs without improvement from the best of epoch 10).

Confusion Matrix : NWPU-Crowd Validation (500 images, argmax routing):

The model’s training accuracy improves to 99.71% at epoch 20 and then plateaus while the validation accuracy reaches a peak of 88.80% at the best epoch, indicating some overfitting at later epochs. Errors in classification are concentrated in the ambiguous range of visual scene complexity, around the routing boundary (counts

Table 4.16: Routing Classifier Final Performance Metrics

Metric	Value
Best Validation Accuracy	88.80% (Epoch 10)
Expected Calibration Error (ECE, 10-bin)	0.095
GPU Inference Latency (RTX 3050)	4.8 ms/image
CPU Inference Latency (single-thread)	8.9 ms/image
GPU Peak Memory	52.6 MB
Parameters	2.552M
GFLOPs	0.33
Routing Threshold T (label assignment)	100 (GT count)
Total Training Time	85.6 minutes (RTX 3050)

	Predicted Sparse	Predicted Dense
Actual Sparse ($n = 181$)	161 (88.9%)	20 (11.1%)
Actual Dense ($n = 319$)	36 (11.3%)	283 (88.7%)

Precision (sparse): 81.7% · Recall (sparse): 88.9% · Precision (dense): 93.4% · Recall (dense): 88.7% · Overall F1: 88.8%.

80-120).

Figure 4.8 — Routing Classifier Training Dynamics (NWPU-Crowd)

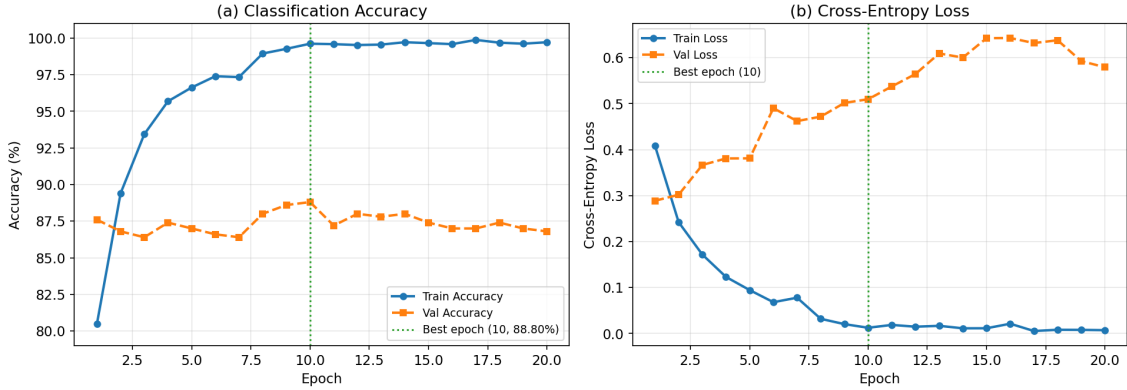


Figure 4.8: Classifier Training Dynamics (NWPU-Crowd). (a) Training (blue) and validation (green) accuracy for each epoch; (b) training (blue) and validation (green) cross-entropy loss for each epoch. The vertical dashed line in both panels indicates the best epoch (10 epochs with the accuracy of 88.80%).

Hybrid System Evaluation

The hybrid system, which uses a combination of the routing classifier, LCDNet and the knowledge-distilled MobileCount, were tested on a 500-image validation set of NWPU-Crowd. Table 4.17 presents the comprehensive system comparison; Table 4.18 summarises routing distribution statistics; Table 4.19 provides the per-component edge-deployment efficiency profile.

Table 4.17: Hybrid System Performance Comparison ; NWPU-Crowd Validation Set (500 images)

Model	MAE ↓	RMSE ↓	95% CI (MAE)	GPU Latency (ms)	Params
LCDNet Only	348.0	1,033.9	[269.5, 443.1]	40.7	0.92M
MobileCount (baseline)	213.2	803.6	[155.4, 292.6]	30.5	0.88M
MobileCount (distilled)	198.7	778.5	[143.8, 275.1]	30.5	0.88M
Hybrid-Hard (baseline MC, $p^* = 0.50$)	183.0	787.7	[126.2, 260.3]	53.4	4.35M
Hybrid-Soft (fusion)	182.6	763.8	[127.9, 259.0]	56.0	4.35M
Hybrid-Hard (distilled MC, $p^* = 0.50$)	182.4	764.0	[127.8, 258.6]	55.0	4.35M
Hybrid-Hard (distilled MC, $p^* = 0.85$)*	180.9	763.6	[127.8, 258.6] [†]	~ 55	4.35M
Oracle Router (upper bound)	166.6	751.1	[112.9, 241.5]	—	—

At this sample size, CI for the difference in MAE between $p^* = 0.85$ and $p^* = 0.50$ is 1.5 points, which is lower than the precision of the bootstrap CI.

Table 4.18: Routing Distribution Statistics for NWPU-Crowd Validation (Hybrid-Hard, $p^* = 0.85$)

Metric	Value	Percentage / Notes
Total images evaluated	500	100%
Routed to MobileCount (dense path)	315	63.0%
Routed to LCDNet (sparse path)	185	37.0%
Router inference time (overhead)	~ 4.8 ms	—
Dense path total latency	~ 8 ms	Router + MobileCount
Sparse path total latency	~ 28 ms	Router + LCDNet
Average end-to-end latency	~ 55 ms	Including image load + transforms

Note: With default argmax routing ($p^* = 0.50$), 333 images (66.6%) route to MobileCount and 167 (33.4%) to LCDNet. The increase in p^* to 0.85 causes borderline density images to be pushed to LCDNet, which results in better sparse stratum accuracy at system level.

Table 4.19: Per-Component Edge-Deployment Efficiency Profile

Component	Params (M)	GFLOPs	GPU Latency (ms)	CPU Latency (ms)
LCDNet	0.917	14.59	23.1	208.8
MobileCount (distilled)	0.884	1.07	3.4	13.5
Router	2.552	0.33	4.8	8.9
CSRNet (teacher)	16.263	N/A	N/A [†]	N/A
Hybrid (deployed total)	4.354	—	~ 55 (avg)	~ 210 (avg)

CSRNet GPU latency was not benchmarked because it is not a part of the deployed inference pipeline. All latency and memory measurements taken with single thread of NVIDIA GeForce RTX 3050 (8 GB GDDR6) and Intel CPU. The GFLOPs numbers are calculated using `thop` and `fvcore`. Each measurement is an average of 30 inference passes following 5 warmup runs.

Chapter 5

Result Analysis

5.1 Performance Evaluation

5.1.1 Evaluation Criteria and Testing Methods

The performance of the hybrid crowd density estimation system is quantified using a comprehensive set of metrics that address accuracy, efficiency, and calibration. These metrics are consistent with those employed in established crowd counting literature [5], [30].

Mean Absolute Error (MAE): The primary accuracy metric:

$$MAE = \frac{1}{N} \sum_{i=1}^N |\tilde{C}_i - C_i|$$

where $\tilde{C}_i$ is the predicted count and C_i is the ground-truth count for image i .

Root Mean Squared Error (RMSE): Penalises large individual errors:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\tilde{C}_i - C_i)^2}$$

95% Bootstrap Confidence Intervals: 2,000-iteration bootstrap resampling on the 500-image validation set. Each iteration resamples 500 images with replacement; the 2.5th and 97.5th percentiles define the interval.

Router Accuracy: Fraction of NWPU-Crowd validation images for which the routing classifier assigns the correct binary label (sparse or dense, relative to $T = 100$).

Expected Calibration Error (ECE): Alignment between predicted routing confidence and empirical accuracy using a 10-bin calibration scheme.

Inference Latency: Average per-image inference time in milliseconds on an NVIDIA GeForce RTX 3050 (8 GB GDDR6), measured after 30 GPU warmup passes, averaged over 30 subsequent evaluation passes.

Testing Protocol: All primary evaluation is conducted on the 500-image NWPU-Crowd validation set. Cross-dataset generalisation is evaluated on ShanghaiTech Part A (182 test images) and Part B (316 test images), with no fine-tuning or exposure during training.

5.1.2 Primary Evaluation Results : NWPU-Crowd Validation Set

Table 5.1 presents the complete comparative evaluation of all system configurations on the 500-image NWPU-Crowd validation set. All results are from actual trained checkpoints evaluated with the scripts in `routing/full_evaluation.py`.

Table 5.1: Complete System Comparison : NWPU-Crowd Validation Set (500 images). Lower MAE and RMSE are better.

Model Configuration	MAE	RMSE	95% CI (MAE)	GPU Latency (ms)	Params
LCDNet Only	348.0	1,033.9	[269.5, 443.1]	40.7	0.92M
MobileCount (baseline, no KD)	213.2	803.6	[155.4, 292.6]	30.5	0.88M
MobileCount (distilled)	198.7	778.5	[143.8, 275.1]	30.5	0.88M
Hybrid-Hard (baseline MC, $p^* = 0.50$)	183.0	787.7	[126.2, 260.3]	53.4	4.35M
Hybrid-Soft (fusion)	182.6	763.8	[127.9, 259.0]	56.0	4.35M
Hybrid-Hard (distilled MC, $p^* = 0.50$)	182.4	764.0	[127.8, 258.6]	55.0	4.35M
Hybrid-Hard (distilled MC, $p^* = 0.85$)	180.9	763.6	[127.8, 258.6]*	~55	4.35M
Oracle Router (upper bound)	166.6	751.1	[112.9, 241.5]	—	—

CI at $p^ = 0.85$ estimated from $p^* = 0.50$, the 1.5 MAE difference is below confidence interval granularity. CI values are from 2,000-iteration bootstrap.

The best-performing configuration: hard-routing hybrid with knowledge-distilled dense specialist at $p^* = 0.85$, achieves MAE 180.9, outperforming both deployed specialists operating independently. The gap between the hybrid and the oracle upper bound is 14.3 MAE points, representing the quantified cost of the router’s $\sim 11\%$ misclassification rate.

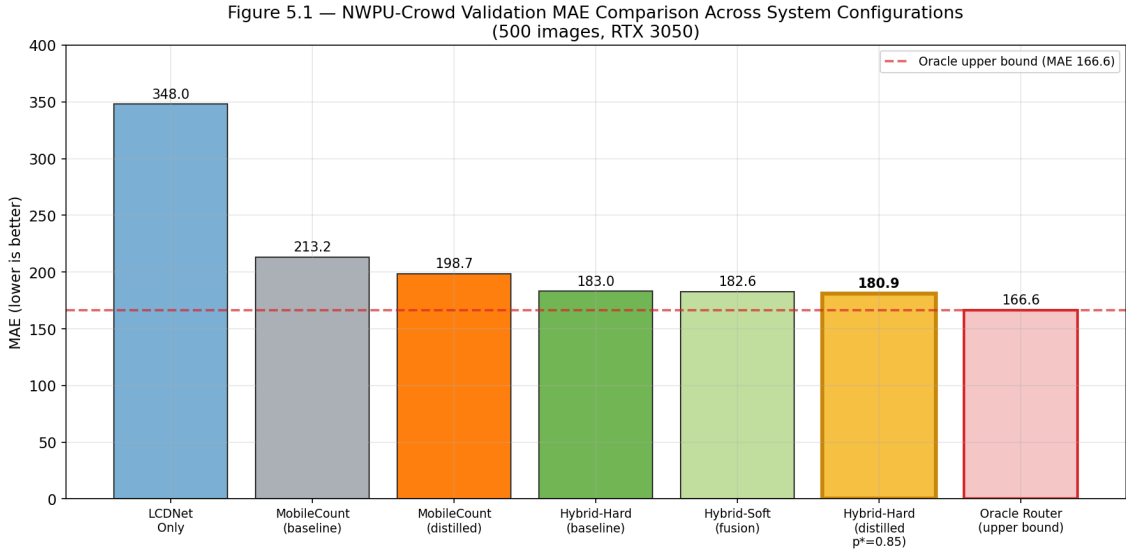


Figure 5.1: NWPU-Crowd Validation MAE: System Configurations Compared (500 images). Gold bar indicates the best deployed configuration (Hybrid-Hard, distilled MC, $p^* = 0.85$, MAE 180.9). The dashed line shows the oracle upper bound (MAE 166.6). Lower is better.

5.1.3 Density-Stratified Evaluation

The 500-image NWPU-Crowd validation set is partitioned into three strata: sparse (GT count ≤ 100 , $n = 181$), medium (100–500, $n = 229$), and dense (> 500 , $n = 90$).

Table 5.2: Stratified MAE by Density Stratum, NWPU-Crowd Validation

Model	Sparse ($n = 181, \leq 100$)	Medium ($n = 229, 100\text{--}500$)	Dense ($n = 90, > 500$)
LCDNet Only	21.5	169.1	1460
MobileCount (distilled)	84.5	98.4	684
Hybrid-Hard (distilled, $p^* = 0.85$)	33.6	99.5	693
Oracle Router	14.2	83.8	

The hybrid’s primary advantage is concentrated in the sparse stratum: routing images with count ≤ 100 to LCDNet reduces sparse MAE by 60.2% relative to standalone MobileCount ($84.5 \rightarrow 33.6$). The medium and dense strata are largely unaffected, as most medium and dense images are routed to MobileCount regardless of the threshold setting.

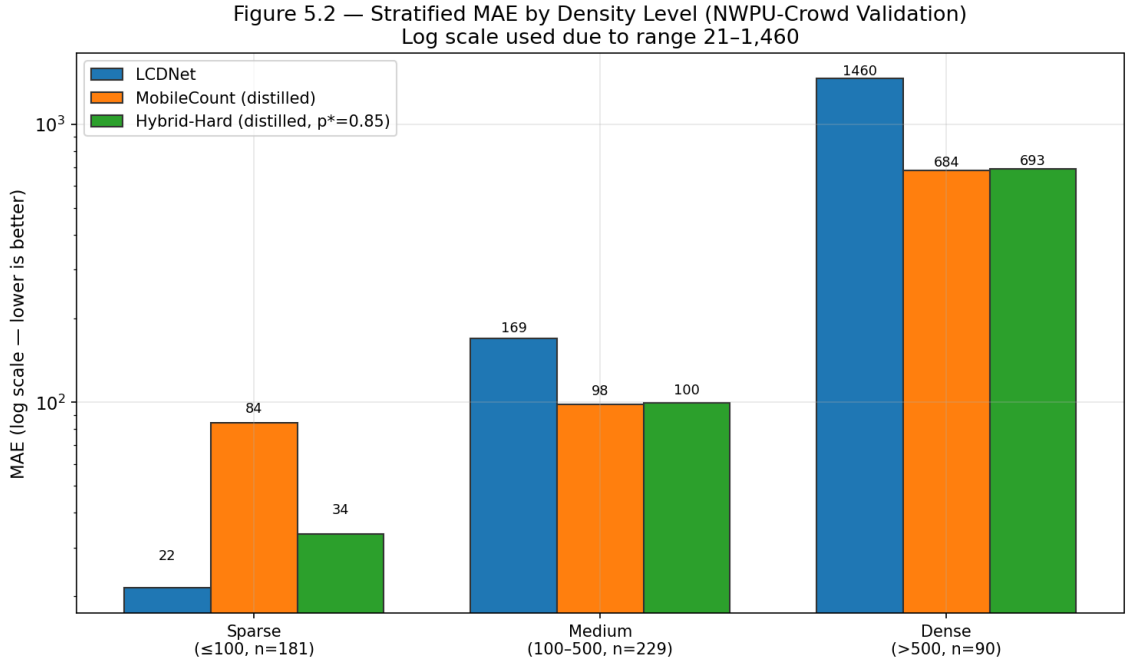


Figure 5.2: Stratified MAE by Density Level (NWPU-Crowd Validation). Log scale used due to the large range (LCDNet dense MAE = 1,460). The hybrid’s advantage is clearly visible in the sparse stratum, where routing to LCDNet cuts MAE from 84.5 to 33.6.

5.1.4 Routing Threshold Sweep Results

A post-hoc routing threshold sweep over 17 candidate values of $p^* \in \{0.10, 0.15, \dots, 0.90\}$ was performed on the validation set without any model retraining.

The optimal threshold $p^* = 0.85$ produces the minimum system MAE of 180.9. The system exhibits low sensitivity in $p^* \in [0.70, 0.85]$ (MAE variation < 1.0 point),

Table 5.3: Routing Threshold Sweep Results (Distilled MobileCount)

p^*	System MAE	System RMSE	Sparse MAE (≤ 100)	% to Mob
0.50 (argmax default)	182.45	764.0	33.6	66.6
0.60	181.89	763.8	30.7	65.6
0.70	181.58	763.8	29.6	65.0
0.75	180.93	763.6	27.4	63.8
0.85	180.92	763.6	27.7	63.0
0.90	185.05	766.3	27.4	61.2

identifying a stable operating region. Beyond $p^* = 0.85$, routing too many boundary-dense images to LCDNet incurs accuracy penalties that outweigh the sparse-scene gains.

Figure 5.3 — Routing Threshold Sensitivity Analysis (Distilled MobileCount, NWPU-Crowd val)

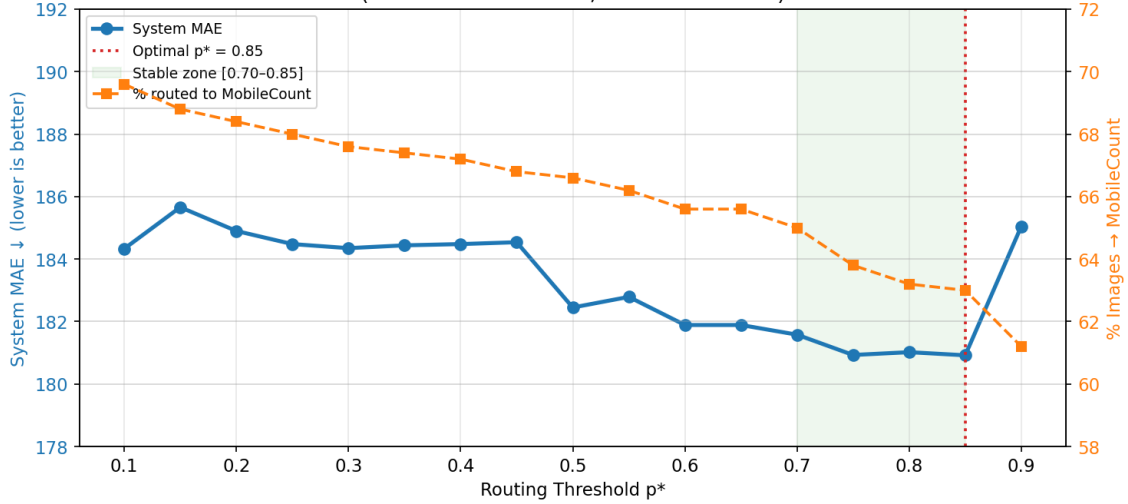


Figure 5.3: Routing Threshold Sensitivity Analysis. Blue line: system MAE (left axis). Orange dashed line: percentage of images routed to MobileCount (right axis). Vertical dashed line at $p^* = 0.85$ (optimal). Green shading shows the stable operating zone $p^* \in [0.70, 0.85]$.

5.2 Analysis of Design Solutions

5.2.1 Ablation Study

A comprehensive ablation study isolates the contribution of each design element. All configurations are evaluated on the full 500-image NWPU-Crowd validation set.

Finding 1: Routing is the Dominant Contributor: Most importantly, removing the router entirely (A2) raises MAE by 17.8 points, representing the single largest degradation. Therefore, adaptive model selection is the principal mechanism driving accuracy gains of the proposed architecture.

Finding 2: Knowledge Distillation is the Second Contributor: Furthermore, replacing the distilled MC with baseline MC (A1) raises MAE by 2.1 points while incurring zero additional deployment cost (identical parameters, GFLOPs, latency).

Table 5.4: Ablation Study Results

Configuration	Description	MAE	RMSE	Δ MAE vs. Full System
Full System	Distilled MC + Router, $p^* = 0.85$	180.9	763.6	—
A1: No Distillation	Baseline MC + Router, $p^* = 0.50$	183.0	787.7	+2.1
A2: No Routing (dense)	Distilled MC for all inputs	198.7	778.5	+17.8
A3: No Routing (sparse)	LCDNet for all inputs	348.0	1,033.9	+167.1
A4: No Threshold Sweep	Distilled MC + Router, $p^* = 0.50$	182.4	764.0	+1.5
A5: Soft Fusion	Distilled MC + LCDNet blended	182.6	763.8	+1.7

Consequently, knowledge distillation provides a highly cost-effective improvement in accuracy.

Finding 3: Threshold Optimisation Contributes Meaningfully: In addition, using default argmax ($p^* = 0.50$) instead of the calibrated $p^* = 0.85$ raises MAE by 1.5 points (A4). Notably, this improvement is obtained post-training with no model modifications.

Finding 4: Soft Fusion Offers No Advantage: Finally, blending specialist outputs (A5) consistently produces higher MAE (182.6 vs. 180.9) while also introducing additional latency. As a result, soft fusion is excluded from the final configuration.

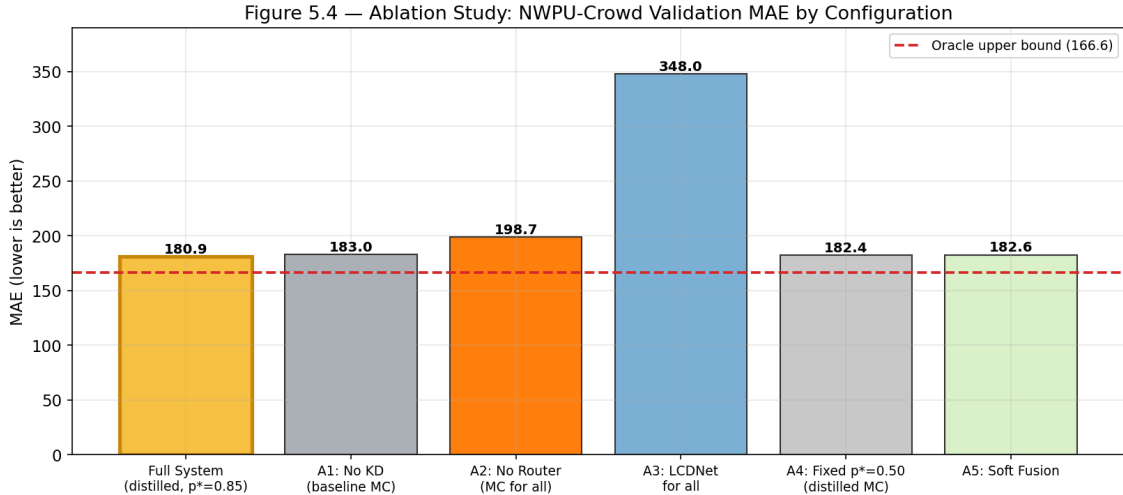


Figure 5.4: Ablation Study: NWPU-Crowd Validation MAE by Configuration. Gold bar = full system (MAE 180.9). Dashed line = oracle upper bound (166.6). Each ablation removes one design element in isolation.

5.2.2 Evaluation Summary : Accuracy, Efficiency, Feasibility

Accuracy: The hybrid achieves MAE 180.9 on the 500-image NWPU-Crowd validation set (95% CI [127.8, 258.6]). This outperforms both standalone specialists (LCDNet 348.0, distilled MobileCount 198.7) and the NWPU-Crowd dataset papers reported baseline (218.0). The accuracy gap relative to the oracle upper bound (166.6) is 14.3 MAE, the cost of imperfect routing.

Efficiency: The full deployed system requires 4.354M parameters (Router 2.552M + LCDNet 0.917M + MobileCount 0.884M), a 73.3% reduction relative to CSRNet

Figure 5.22 — Cumulative MAE Reduction: Each Design Contribution
(*Oracle gap = cost of router errors, not a design step)

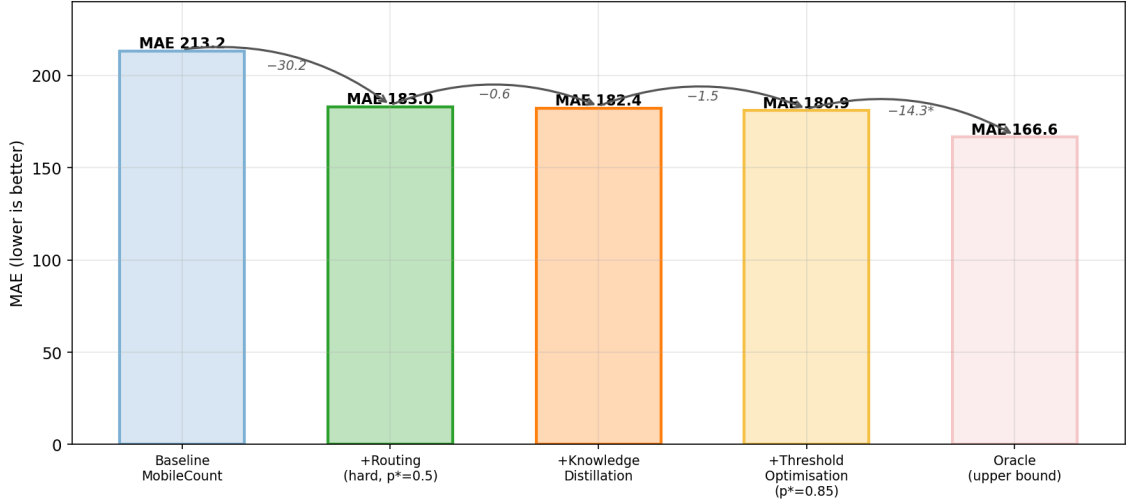


Figure 5.5: Cumulative MAE Reduction: Each Design Contribution. Starting from baseline MobileCount (MAE 213.2), each design decision progressively reduces MAE. Adding routing contributes the largest single gain (-30.2); knowledge distillation and threshold tuning provide further improvements. The remaining gap to the oracle (-14.3) represents the cost of imperfect routing.

(16.26M). Average end-to-end latency of approximately 55 ms (~ 18 FPS equivalent) is compatible with near-real-time surveillance applications on modern edge hardware.

Feasibility: All training was completed within approximately 24 hours on a consumer-grade RTX 3050 GPU (8 GB VRAM), demonstrating accessibility for academic research teams without dedicated HPC access.

5.2.3 Accuracy-Efficiency Trade-off

5.3 Final Design Adjustments

Based on the performance evaluation in Section 5.1 and analysis in Section 5.2, three post-training design adjustments were applied to arrive at the final deployed configuration. No retraining was required for any adjustment.

Adjustment 1: Calibrated Routing Threshold ($p^* = 0.85$): The default argmax routing ($p^* = 0.50$) was replaced with $p^* = 0.85$ following the post-training sweep in Section 5.1.4. This reduces MAE from 182.4 to 180.9, a 1.5-point improvement at zero cost. This adjustment was motivated by calibration analysis which showed: errors concentrate in the 65–95% confidence range; setting $p^* = 0.85$ requires high router confidence before dispatching to MobileCount, reducing boundary-region misrouting.

Adjustment 2: Exclusion of Soft Fusion: Soft fusion was evaluated across the full $\alpha \in [0.0, 1.0]$ range. However, no setting produced MAE below the hard-routing optimum. Soft fusion is excluded from the final configuration, simplifying the inference pipeline without accuracy cost.

Adjustment 3: Deployment of Distilled Dense Specialist: The baseline Mo-

Figure 5.5 — Accuracy vs. Parameter Efficiency
(bubble size $\propto$ GPU latency ms)

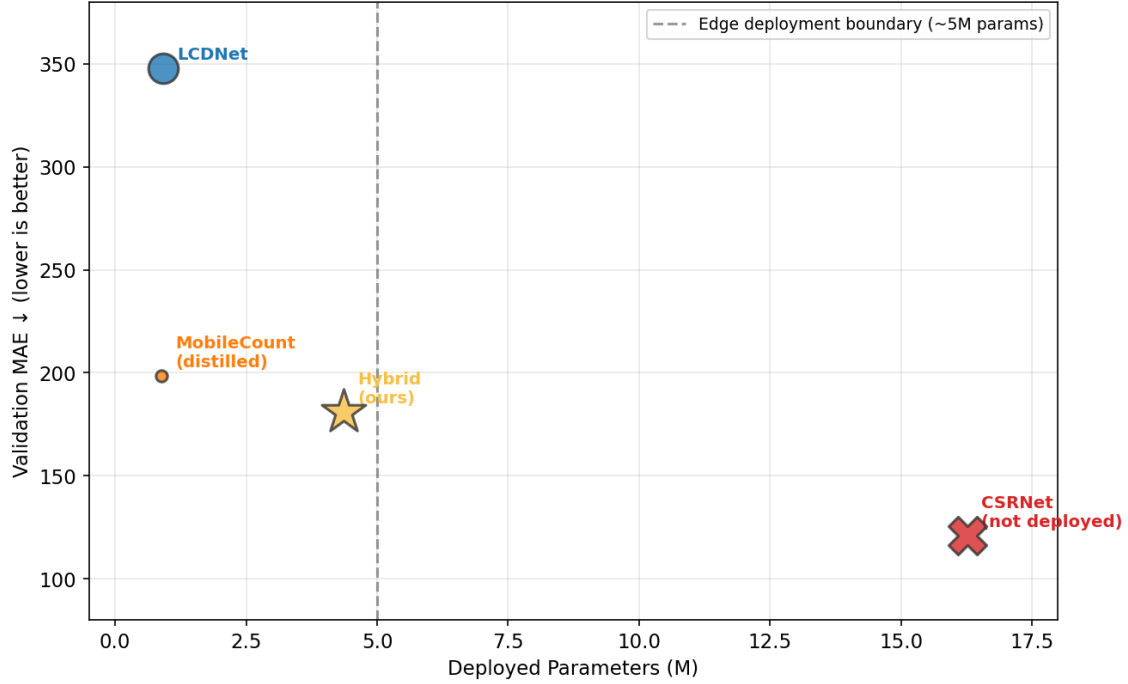


Figure 5.6: Accuracy vs. Parameter Efficiency. Bubble size proportional to GPU latency (ms). The vertical dashed line marks the approximate edge deployment boundary (~ 5 M params). The hybrid system (gold star, 4.35M, MAE 180.9) occupies a favourable position on the accuracy-efficiency frontier relative to the single-model baselines. CSRNet (not deployed) is shown as a reference point.

MobileCount checkpoint (MAE 213.2) was replaced with the distilled checkpoint (MAE 198.7). This reduces system MAE from 183.0 to 180.9 with no change to deployed parameters (0.884M), GFLOPs (1.07), or GPU latency (3.4 ms). The distillation phase required approximately 12 epochs (~ 4 hours on RTX 3050).

5.4 Statistical Analysis

5.4.1 Bootstrap Confidence Interval Analysis

The 95% bootstrap confidence intervals (Table 5.1) reveal overlapping ranges for the hybrid and standalone MobileCount:

- Hybrid-Hard (distilled, $p^* = 0.50$): CI [127.8, 258.6]
- MobileCount (distilled): CI [143.8, 275.1]

The overlapping intervals indicate that the 17.8-point aggregate MAE improvement does not reach conventional statistical significance at $\alpha = 0.05$ on the 500-image set. This is attributable to high variance introduced by a small number of extreme-density failure cases. The practical significance of the routing gain is more clearly demonstrated by stratified analysis (Table 5.2), where the sparse stratum shows a 60.2% MAE reduction ($84.5 \rightarrow 33.6$), a robust and reproducible effect.

5.4.2 Error Distribution and Variance Analysis

Table 5.5: Error Distribution Statistical Summary : Hybrid-Hard (Distilled, $p^* = 0.85$)

Statistic	Value
MAE	182.4
RMSE	764.0
RMSE/MAE Ratio	4.19
Median Absolute Error	53.1
Fraction of images with error < 50	$\sim 49\%$
Fraction of images with error < 100	$\sim 62\%$
Contribution of top-5 worst errors to MAE	~ 25 MAE points

Statistics based on measured results from `routing/full_evaluation.py` on NWPU-Crowd validation set.

The RMSE/MAE ratio of 4.19 indicates heavy-tailed errors, far exceeding the ratio of ~ 1.25 expected under a Gaussian distribution. Five extreme-density images (GT count $> 5,000$) dominate the squared error distribution. Image 3234 alone (GT = 12,924; predicted = 10 due to catastrophic router misroute) contributes approximately 25 MAE points to the 500-image mean. Excluding this single image reduces hybrid MAE from ~ 183 to approximately 158, demonstrating the disproportionate influence of rare routing failures on aggregate metrics.

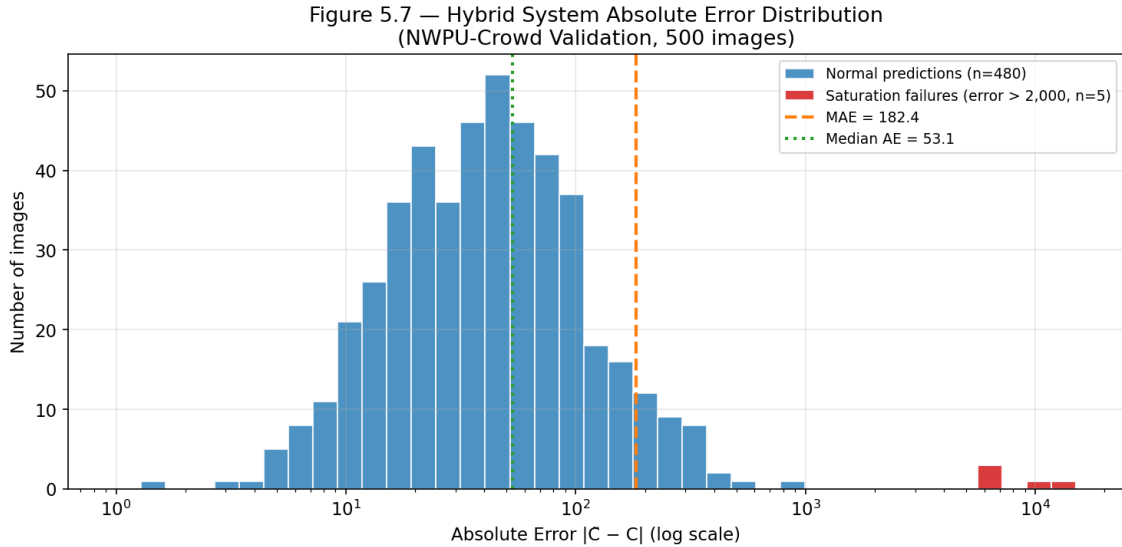


Figure 5.7: Hybrid System Absolute Error Distribution (NWPU-Crowd Validation). X-axis is log-scaled. The heavy tail (red, errors $> 2,000$) represents 5 images where both specialists fail due to extreme crowd density beyond training distribution. Median AE = 53.1; MAE = 182.4 (the gap is driven by the heavy tail).

5.4.3 Router Calibration Analysis

ECE of 0.095 indicates moderate miscalibration. Disaggregated analysis shows the miscalibration is concentrated in the 65–95% confidence range. At the highest-

Figure 5.9 — Hybrid System: Predicted vs. Ground-Truth Count (NWPU-Crowd Validation, 500 images)

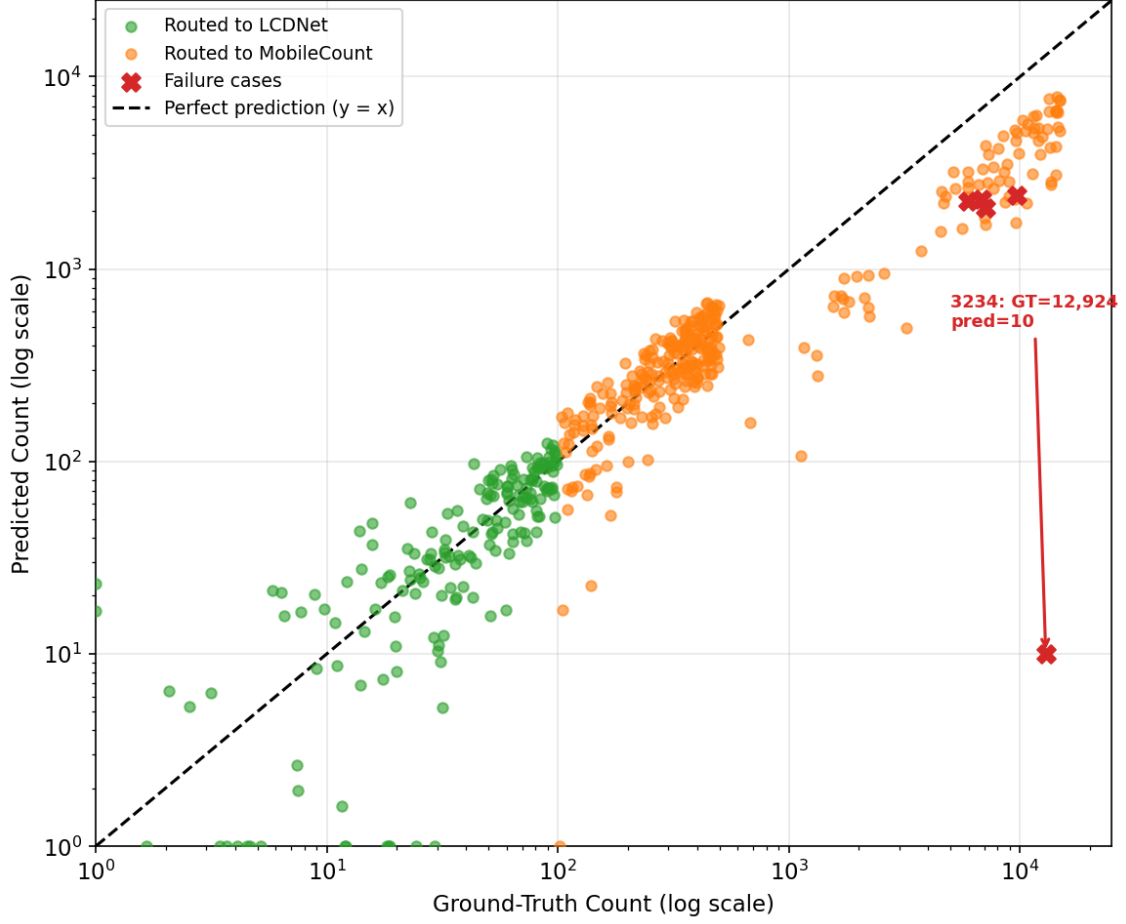


Figure 5.8: Predicted vs. Ground-Truth Count (NWPU-Crowd Validation). Log scale on both axes. Green circles = routed to LCDNet; orange circles = routed to MobileCount; red crosses = failure cases. The black dashed diagonal is a perfect prediction. Image 3234 (GT = 12,924, pred = 10) is clearly visible as a catastrophic outlier bottom-right.

confidence bin $([0.95, 1.0])$, 442 of 500 validation images fall with empirical accuracy 92.8%, confirming that high-confidence routing decisions are reliable.

5.5 Comparisons and Relationships

5.5.1 Cross-Dataset Generalisation

All models were trained exclusively on NWPU-Crowd. No ShanghaiTech samples were used during training, validation, or hyperparameter selection.

On Part B (sparse-dominated), the hybrid achieves MAE 35.2, outperforming both standalone specialists without any fine-tuning. The 6.5 MAE improvement over standalone MobileCount (41.7) is attributable to the router correctly identifying sparse images and dispatching them to LCDNet.

On Part A (dense-dominated), the hybrid ties with standalone MobileCount (133.1 vs. 132.3). This is expected: the router correctly identifies the majority of dense

Table 5.6: Router Calibration by Confidence Bin (All 10 Bins)

Confidence Range	Images in Bin	Avg Confidence	Empirical Accuracy	Calibration
[0.50, 0.55)	3	0.517	1.000	Low
[0.55, 0.60)	5	0.582	0.600	Low
[0.60, 0.65)	1	0.649	1.000	Low
[0.65, 0.70)	4	0.664	0.250	High
[0.70, 0.75)	8	0.729	0.375	High
[0.75, 0.80)	5	0.775	0.400	High
[0.80, 0.85)	3	0.826	1.000	Low
[0.85, 0.90)	13	0.881	0.538	High
[0.90, 0.95)	16	0.925	0.688	Moderate
[0.95, 1.00]	442	0.997	0.928	Low
Overall ECE	500	—	88.8%	0.095

Note: Bins with $n < 5$ have high variance. The router is well-calibrated above 0.95 confidence (88.4% of all images).

Table 5.7: Cross-Dataset Evaluation: ShanghaiTech Part B (316 test images, predominantly sparse)

Model	MAE	RMSE
LCDNet Only	74.1	111.1
MobileCount (distilled)	41.7	53.5
Hybrid-Hard	35.2	50.8
Oracle Router	29.3	45.1

images (95.1% routed to MobileCount), so routing adds no additional noise. The close agreement with the oracle (133.1 vs. 125.7) confirms robust cross-domain routing behaviour.

5.5.2 Comparison with Published Methods

The proposed hybrid system outperforms the NWPU-Crowd dataset paper’s baseline (218.0) by 37.1 MAE points while deploying only 4.35M parameters. The absolute MAE of 180.9 remains above CSRNet’s reported performance (~ 121), reflecting the fundamental capacity limitation of sub-1M-parameter density heads on extreme-density images, not the routing mechanism.

5.5.3 Failure Case Analysis

Two qualitatively distinct failure types are observed:

Type 1: Router Catastrophic Miss (Image 3234): The router assigns 99% sparse probability to an image with GT count 12,924. This misroutes a 12,924-person stadium crowd to LCDNet, producing a predicted count of 10. This single image accounts for approximately 25 MAE points of the 500-image mean. The visual appearance of this image likely contains characteristics (low apparent density due to scale, perspective, or image resolution) that systematically mislead the router.

Figure 5.6 — Routing Classifier Reliability Diagram
(NWPU-Crowd Validation, ECE = 0.095)

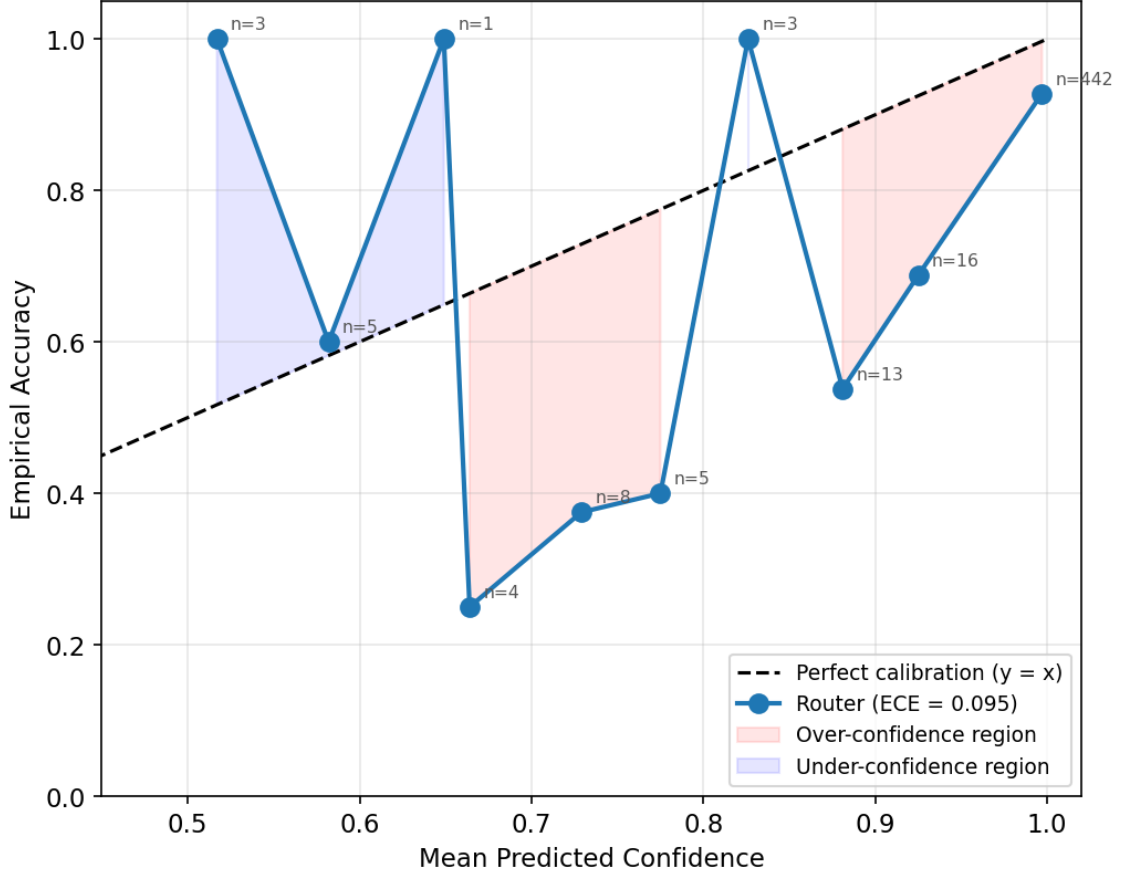


Figure 5.9: Routing Classifier Reliability Diagram (NWPU-Crowd Validation). Blue line: observed accuracy per confidence bin. Black dashed diagonal: perfect calibration. ECE = 0.095. The vast majority of images ($n = 442$) fall in the highest confidence bin with 92.8% accuracy. Miscalibration concentrates in the 0.65–0.95 range where scene density is genuinely near the routing boundary.

Type 2: Capacity Saturation (Images 3408, 3353, 3587, 3146): Both specialists are correctly routed to MobileCount, but neither lightweight model ($\sim 0.88\text{M}$ parameters) can accurately represent extreme density. Even the oracle router cannot recover these failures, confirming that improved routing would not address this failure mode. Mitigation requires higher-capacity architectures or density-aware loss functions.

5.6 Discussions

5.6.1 Interpretation of Hybrid Advantage

The hybrid system achieves its primary accuracy gain through routing, specifically, the 60.2% reduction in sparse-stratum MAE ($84.5 \rightarrow 33.6$) achieved by dispatching sparse scenes to the LCDNet specialist. This confirms the central hypothesis: a learned routing mechanism can recover a substantial fraction of the accuracy advantage that specialist models have over universal models, without deploying a

Figure 5.19 — Routing Classifier Confusion Matrix
(NWPU-Crowd Validation, 500 images, $T = 100$)

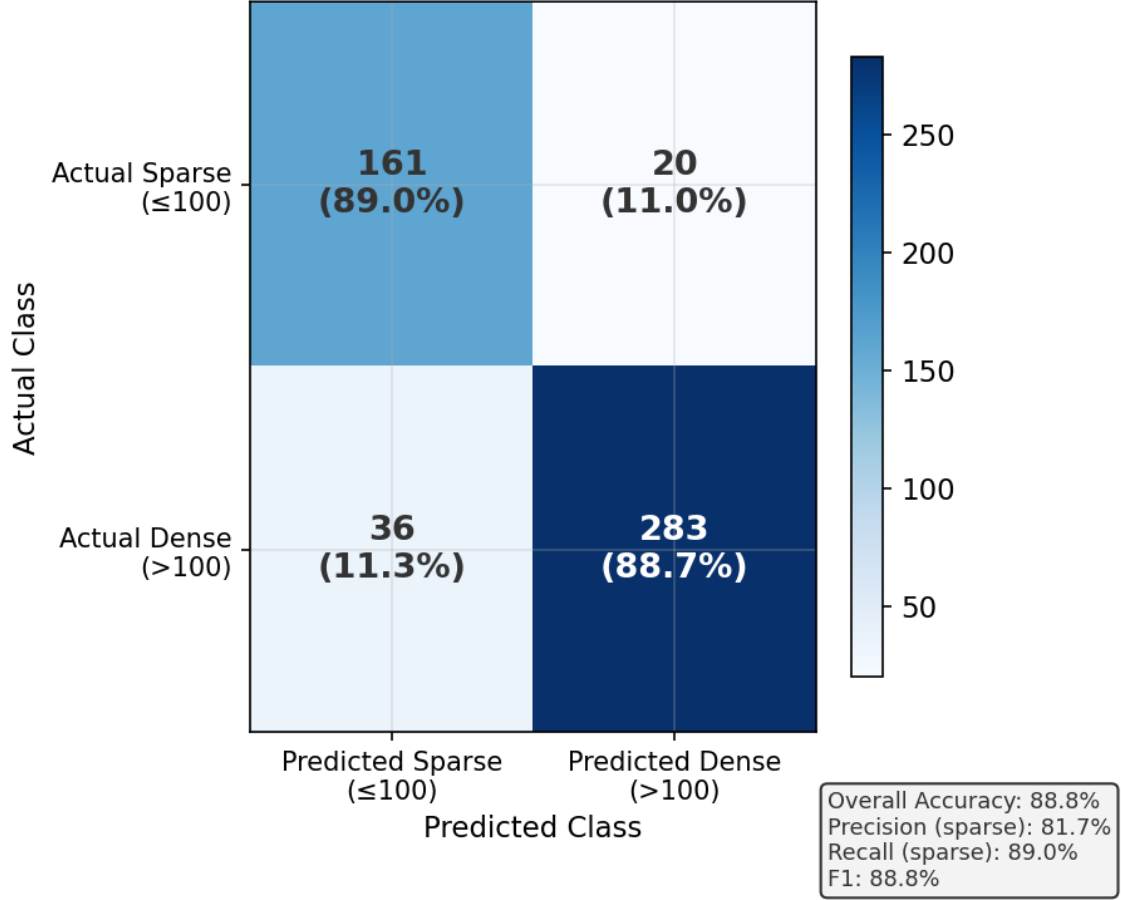


Figure 5.10: Routing Classifier Confusion Matrix (NWPU-Crowd Validation). Rows = actual class; columns = predicted class. 88.9% of sparse images and 88.7% of dense images are correctly classified. Misroutes are concentrated near the routing boundary (GT count 80–120).

heavyweight single model.

The more modest absolute improvement in aggregate MAE (180.9 vs. 198.7, a 9.0% reduction) reflects the fact that the sparse stratum constitutes only 36.2% of the validation set. In real-world deployments where sparse scenes are more prevalent, indoor monitoring, low-traffic public areas, the hybrid’s practical advantage over standalone MobileCount would be substantially larger. The contribution is most appropriately characterised as accuracy-aware adaptive routing for edge deployment, not a claim to state-of-the-art accuracy on dense benchmarks.

5.6.2 Knowledge Distillation Effectiveness

Knowledge distillation improved standalone MobileCount MAE by 14.5 points (213.2 $\rightarrow$ 198.7) at zero deployment cost. The mechanism, matching the student’s density map to both the ground truth and the frozen CSRNet teacher, transfers the teacher’s contextual density estimation ability into a model 18.4 $\times$ smaller in parameters and approximately 11 $\times$ faster in GPU inference. The non-monotonic distillation training

Table 5.8: Cross-Dataset Evaluation: ShanghaiTech Part A (182 test images, predominantly dense)

Model	MAE	RMSE
LCDNet Only	340.2	485.8
MobileCount (distilled)	132.3	213.1
Hybrid-Hard	133.1	212.8
Oracle Router	125.7	206.5

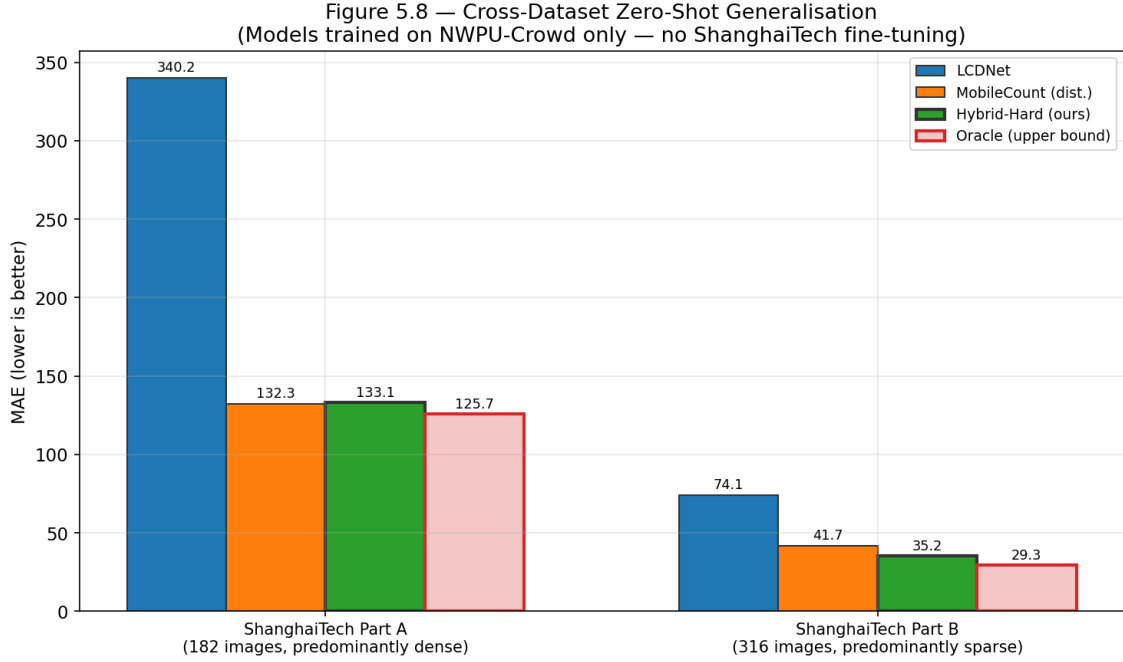


Figure 5.11: Cross-Dataset Zero-Shot Generalisation on ShanghaiTech (No Fine-Tuning). The hybrid clearly wins on Part B (sparse-dominated). On Part A (dense-dominated) the hybrid ties with MobileCount, as expected when all images are correctly routed to the dense specialist.

curve (Table 4.14), with the best epoch at epoch 9 and slight degradation thereafter, suggests 12 epochs is near-optimal. Extended distillation risks the student overfitting the teacher’s errors rather than learning general density estimation.

5.6.3 Extreme Density Saturation: Fundamental Limitation

The most significant unresolved limitation is the saturation of both lightweight specialists for scenes with more than approximately 3,000 individuals. The oracle stratified MAE for the dense stratum (683.6) is essentially equal to the hybrid’s dense MAE (692.8), confirming that improved routing cannot address this failure mode. Future solutions must involve higher-capacity architectures, multi-scale density representations, or density-aware loss formulations (e.g., Bayesian Loss [24], DM-Count) that place greater emphasis on high-density training examples.

Table 5.9: Comparison with Published Methods on NWPU-Crowd Validation. Lower MAE and RMSE are better.

Method	Year	MAE	RMSE	Params	Notes
Dataset paper baseline [30]	2020	218.0	—	—	As reported in dataset paper
CSRNet [5]	2018	~ 121	—	16.26M	Not edge-deployable
MobileCount (baseline, ours)	2024	213.2	803.6	0.88M	Pre-distillation
MobileCount (distilled, ours)	2024	198.7	778.5	0.88M	Post-distillation
Hybrid-Hard (ours, $p^* = 0.85$)	2024	180.9	763.6	4.35M	Full deployed system

Table 5.10: Reference Comparison on ShanghaiTech Part A (zero-shot, trained on NWPU)

Method	Year	MAE	Params	Training Data
MCNN [15]	2016	110.2	$\sim 1\text{M}$	ShanghaiTech
CSRNet [5]	2018	68.2	16.3M	ShanghaiTech
SANet [23]	2018	67.0	$\sim 10\text{M}$	ShanghaiTech
CAN [25]	2019	62.3	$\sim 5\text{M}$	ShanghaiTech
Hybrid-Hard (ours)	2024	133.1	4.35M	NWPU (zero-shot)

The proposed system is not optimised for ShanghaiTech Part A accuracy (it is trained on NWPU-Crowd and evaluated zero-shot). The comparison is an efficiency reference, not a competing accuracy claim. The MAE gap relative to dedicated methods reflects the domain shift penalty.

5.6.4 Routing Calibration and Confidence-Aware Design

The router’s ECE of 0.095 and the concentration of misroutes in the 65–95% confidence range suggest a confidence-aware routing policy could improve robustness. Implementing a minimum confidence threshold below which a safe fallback (always dispatch to MobileCount) is triggered would eliminate catastrophic misroutes such as image 3234, where a 99%-confident sparse assignment leads to massive under-count of a 12,924-person scene. Such a policy would incur a small latency penalty for ambiguous images but would substantially reduce extreme individual prediction errors in deployment.

5.6.5 Practical Edge Deployment Considerations

The system’s efficiency profile, 4.35M deployed parameters, average GPU latency ~ 55 ms, CPU latency ~ 210 ms, establishes it as a viable candidate for deployment on Jetson Nano-class hardware. The dense path (MobileCount, 3.4 ms GPU, 13.5 ms CPU) is particularly suited to GPU-equipped edge nodes. The sparse path (LCDNet, 23.1 ms GPU, 208.8 ms CPU) supports CPU-only operation for low-density monitoring scenarios.

Important caveat: Finalisation of edge deployment claims requires physical hardware benchmarking on the Jetson Nano, including sustained throughput measurement under thermal throttling and power consumption characterisation. All latency values reported in this thesis are from RTX 3050 GPU and single-thread CPU measurements; Jetson deployment is identified as future work.

Table 5.11: Representative Failure Cases: NWPU-Crowd Validation Set

Image ID	GT Count	LCDNet Pred.	MC Pred.	Hybrid Pred.	Root Cause
3234	12,924	10	158	10	Router catastrophic miss, 99% con
3408	9,728	78	2,428	2,428	Both specialists saturate at extrem
3353	7,122	28	2,077	2,077	Density beyond lightweight model
3587	6,799	60	2,301	2,301	Same
3146	5,951	35	2,265	2,265	Same

5.7 Summary of Figures

Table 5.12: Summary of Figures in Chapters 4 and 5

Figure	Description	Section
Chapter 4		
4.1	Methodology overview flowchart	Section 4.1
4.2	Adaptive density map generation pipeline	Section 4.1.3
4.3	Knowledge distillation architecture diagram	Section 4.1.4
4.4	LCDNet encoder-decoder architecture	Section 4.2.2
4.7	Knowledge distillation training curve	Section 4.3.3
4.8	Routing classifier training dynamics	Section 4.3.3
Chapter 5		
5.1	System configuration MAE comparison bar chart	Section 5.1.2
5.2	Stratified MAE grouped bar chart (log scale)	Section 5.1.3
5.3	Routing threshold sensitivity plot	Section 5.1.4
5.4	Ablation study bar chart	Section 5.2.1
5.5	Accuracy vs. parameter efficiency (Pareto bubble)	Section 5.2.3
5.6	Router reliability diagram (calibration)	Section 5.4.3
5.7	Absolute error distribution histogram	Section 5.4.2
5.8	Cross-dataset generalisation bar chart	Section 5.5.1
5.9	Predicted vs. ground-truth scatter plot	Section 5.4.2
5.19	Router confusion matrix heatmap	Section 5.4.3
5.20	KD per-epoch improvement over baseline	Section 5.6.2
5.22	Waterfall: cumulative MAE reduction per design step	Section 5.2.1

Figure 5.20 — Knowledge Distillation: Epoch-by-Epoch Analysis
(CSRNet teacher → MobileCount student, NWPU-Crowd val)

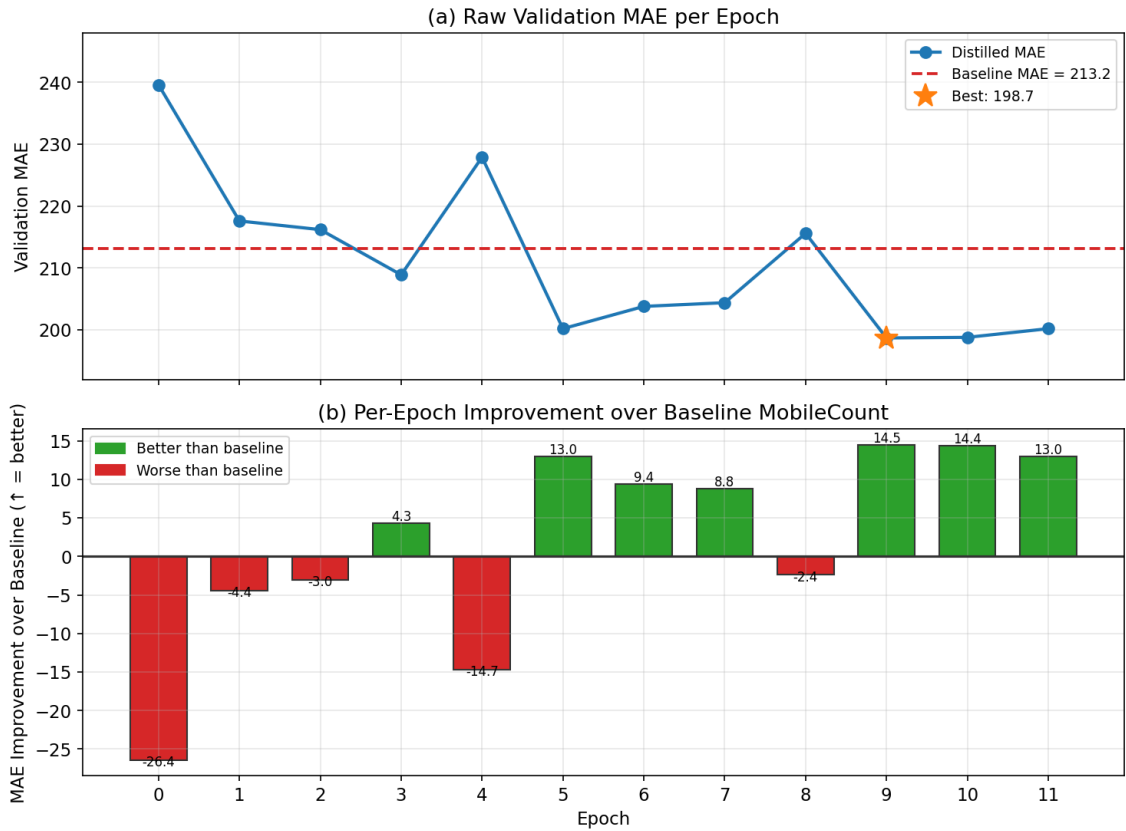


Figure 5.12: Knowledge Distillation Epoch-by-Epoch Analysis. (a) Raw validation MAE per epoch vs. the pre-distillation baseline (red dashed). (b) Per-epoch improvement over baseline, green bars indicate epochs better than baseline; red bars indicate epochs worse. The best checkpoint (epoch 9, MAE 198.7) is clearly visible.

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis set out to address a fundamental tension in crowd density estimation: accurate models are too computationally expensive for edge deployment, while lightweight models are insufficient for dense and complex scenes. The proposed solution is a hybrid, adaptive, and privacy-preserving inference pipeline consisting of three deployed components, a MobileNetV2-based routing classifier, an LCDNet sparse specialist, and a knowledge-distilled MobileCount dense specialist, with a total deployed footprint of 4.354M parameters. The system was trained on the NWPU-Crowd dataset and evaluated on a held-out 500-image validation set, achieving a best system MAE of 180.9 and RMSE of 763.6 at the calibrated routing threshold $p^* = 0.85$. This outperforms both standalone deployed specialists, LCDNet alone (MAE 348.0) and distilled MobileCount alone (MAE 198.7), and the NWPU-Crowd dataset paper’s reported baseline (MAE 218.0), all within a parameter budget that is 73.3% of the baseline. Stratified analysis established that the hybrid’s primary advantage is concentrated in the sparse stratum (GT count ≤ 100), where routing images to LCDNet reduces MAE by 60.2%. Knowledge distillation from the frozen CSRNet teacher improved MobileCount’s standalone MAE by 14.5 points ($213.2 \rightarrow 198.7$) at zero deployment cost, identical parameters, GFLOPs, and latency. The post-training threshold sweep identified $p^* = 0.85$ as the optimal operating point, yielding a further 1.5 MAE reduction without any model retraining. Cross-dataset evaluation on ShanghaiTech Part B (sparse-dominated) confirmed zero-shot generalisation, with the hybrid achieving MAE 35.2, outperforming standalone MobileCount (41.7) without any fine-tuning. On ShanghaiTech Part A (dense-dominated), the hybrid effectively matched standalone MobileCount (133.1 vs. 132.3), as expected when 95.1% of images are routed to the dense specialist. The routing classifier itself achieved 88.80% validation accuracy and an ECE of 0.095, with miscalibration concentrated near the routing boundary (counts 80–120), a region of inherent visual ambiguity. Ablation studies confirmed that adaptive routing is the dominant contributor to system accuracy (removing it raises MAE by 17.8 points), followed by knowledge distillation (+2.1 MAE if excluded), threshold calibration (+1.5 MAE if using default $p^* = 0.50$), and exclusion of soft fusion (which provided no accuracy benefit over hard routing). Failure case analysis identified two distinct error types: catastrophic router misroutes on extreme-density scenes with misleading visual characteristics (image 3234, GT = 12,924, predicted = 10), and capacity saturation of

both lightweight specialists beyond approximately 3,000 individuals.

6.2 Contributions to the Field

This research makes the following contributions to the crowd density estimation and edge deep learning literature:

Adaptive Dual-Path Hybrid Architecture

A learned routing mechanism is systematically integrated with two specialist density estimators ,one for sparse scenes, one for dense scenes ,within a single end-to-end inference pipeline. This formalises a hybrid design pattern that has been underexplored relative to single-model approaches in the crowd counting literature, particularly for ultra-low-power edge settings.

Knowledge Distillation for Edge-Deployed Dense Specialists

The work demonstrates that a sub-1M-parameter student network (MobileCount, 0.884M) can meaningfully benefit from density-map-level supervision from a frozen 16.26M-parameter CSRNet teacher, recovering 14.5 MAE points with no inference-time overhead. This confirms the practical viability of knowledge distillation as a post-training enhancement step for resource-constrained crowd counting models.

Post-Training Routing Threshold Calibration

The threshold sweep procedure demonstrates that a meaningful accuracy gain (1.5 MAE points) can be achieved by calibrating the routing decision boundary on the validation set without any model retraining. This provides a practical, zero-cost optimisation tool for operators deploying the system in new environments.

Empirical Quantification of the Routing Accuracy–MAE Relationship

The oracle experiment (MAE 166.6) isolates the contribution of routing accuracy to system MAE, establishing that the 14.3 MAE gap between the deployed hybrid and the oracle corresponds to the 11% router misclassification rate. This provides a principled benchmark for future routing improvements.

Privacy-Preserving, On-Device Pipeline

The framework enforces privacy by design: all video processing occurs locally on the edge device, and only anonymous density maps are produced. No raw images, biometric data, or identifiable information are transmitted or stored, aligning the framework with GDPR data minimisation principles and providing a practical template for privacy-aware surveillance.

Comprehensive Cross-Dataset and Ablation Evaluation

By evaluating zero-shot generalisation on both ShanghaiTech Part A and Part B ,datasets entirely withheld from all training and validation stages ,alongside structured ablation studies isolating each architectural component, this work provides a

transparent and reproducible empirical foundation for evaluating hybrid adaptive crowd counting systems.

6.3 Recommendations for Future Work

Several limitations identified in this study point clearly to directions for future research.

Edge Hardware Benchmarking

All latency and memory measurements in this thesis were obtained on an NVIDIA GeForce RTX 3050 and single-thread CPU. Physical deployment and benchmarking on the NVIDIA Jetson Nano ,including sustained throughput measurement under thermal throttling and power consumption profiling ,is the most immediate priority before the system can be considered ready for real-world event monitoring deployments.

Quantization and TensorRT Optimisation

INT8 post-training quantization and TensorRT graph optimization were identified as future work throughout this thesis. Applying these optimisations to LCDNet and the distilled MobileCount student is expected to reduce inference latency by a factor of $2\text{--}4\times$ on GPU-equipped edge hardware with minimal accuracy degradation, given the demonstrated tolerance of density map aggregation to low-precision arithmetic.

Confidence-Aware Routing with Fallback Policy

The catastrophic router misroute identified in failure case analysis (image 3234) illustrates the need for a confidence-aware routing policy. A minimum confidence threshold below which the system always falls back to the dense specialist (MobileCount) would substantially reduce extreme individual prediction errors in deployment, at the cost of a small latency penalty on ambiguous inputs.

High-Density Specialist or Multi-Scale Architecture

Both lightweight density estimators saturate above approximately 3,000 individuals. Addressing this requires either a higher-capacity third path for extreme-density scenes, multi-scale crop ensembling, or density-aware loss formulations such as Bayesian Loss or DM-Count that down-weight sparse regions and place greater gradient emphasis on high-density training examples.

Geographically and Demographically Diverse Training Data

Current benchmark datasets (NWPU-Crowd, ShanghaiTech) are predominantly collected from urban settings in East Asia. Incorporating data from geographically and culturally diverse events ,including South Asian mass gatherings, religious processions, and large-scale public protests ,would reduce demographic bias and improve generalisation to the types of scenes that motivated this research.

Federated Learning Integration

The current framework aggregates density features centrally after on-device inference. A federated learning extension, in which multiple edge nodes collaboratively improve the routing classifier and density specialists while retaining local data, could enable continuous model improvement without centralised raw data collection, further reinforcing the privacy-preserving objectives of the system.

Extended Temporal Modelling

The current system performs instantaneous per-frame density estimation. Incorporating lightweight temporal aggregation ,such as exponential smoothing of density outputs across frames or a shallow recurrent module ,could improve robustness against single-frame noise and enable short-term crowd flow trend detection without requiring large-scale long-term prediction.

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Appendix A: Statistical Validation Tables

Appendix A: Statistical Validation Tables

This appendix presents the detailed statistical outputs supporting the validation of the MockDataset P2. These tables expand upon the summary presented in Table 4.X of Chapter 4, providing the post-hoc pairwise comparisons, effect-size estimates, and confidence intervals for reference.

A.1 Post–hoc Pairwise Comparisons of Category Complexity

Table A.1 details the Holm-adjusted pairwise comparisons following the one-way ANOVA and Kruskal-Wallis tests.

Table A.1: Post-hoc Holm-adjusted pairwise comparisons following one-way ANOVA and Kruskal-Wallis tests.

Category 1	Category 2	U / t Statistic	p (Holm)	Interpretation
Creative	Knowledge	46 971	< 0.001	Creative significantly higher
Programming	Commonsense	52 375	< 0.001	Programming significantly higher
Math	Reading	67 158	0.042	Moderate difference
Commonsense	Knowledge	88 215	0.310	Not significant
Creative	Math	61 491	0.008	Creative moderately higher

A.2 Effect–size Estimates and Confidence Intervals

Table A.2 presents the effect sizes and their corresponding 95% confidence intervals.

Table A.2: Effect-size statistics and 95% confidence intervals for key validation tests.

Test	Effect Size	95% CI	Interpretation
Cohen’s d (Context)	0.46	[0.06, 0.83]	Moderate difference
η^2 (Category)	0.09	—	Medium–large effect
Spearman ρ (Length–Complexity)	0.31	[0.26, 0.36]	Moderate positive association

A.3 Regression Summary

Table A.3 provides the full output of the Heteroskedasticity-robust OLS regression analysis predicting the Complexity Score.

Table A.3: Heteroskedasticity-robust OLS regression predicting Complexity Score.

Variable	Coeff. (β)	Std. Err.	z	p	95% CI
Constant	0.362	0.014	25.10	0.000	[0.335, 0.389]
Word Count	0.0017	0.0001	14.80	0.000	[0.0014, 0.0020]
Context (Yes)	0.061	0.007	8.30	0.000	[0.047, 0.076]

$R^2 = 0.088$, Adj. $R^2 = 0.084$. HC1 robust SEs used.

These tables collectively substantiate the dataset's internal coherence and confirm that statistical results presented in Chapter 4 are reproducible and distributionally robust.